

Lecture 46

# Jellyfish I: neuroanatomy of *Cassiopea*, distributed nerve nets, rhopalia, and pacemaker control of rhythmic bell contractions

Date TBD

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## B Upside-down jellyfish *Cassiopea*



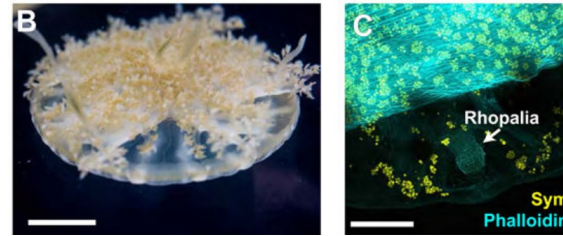
Abrams et al. (2025), Fig. 1(B). *Cassiopea* in its natural upside-down orientation.

# Cassiopea contracts its bell while remaining in place

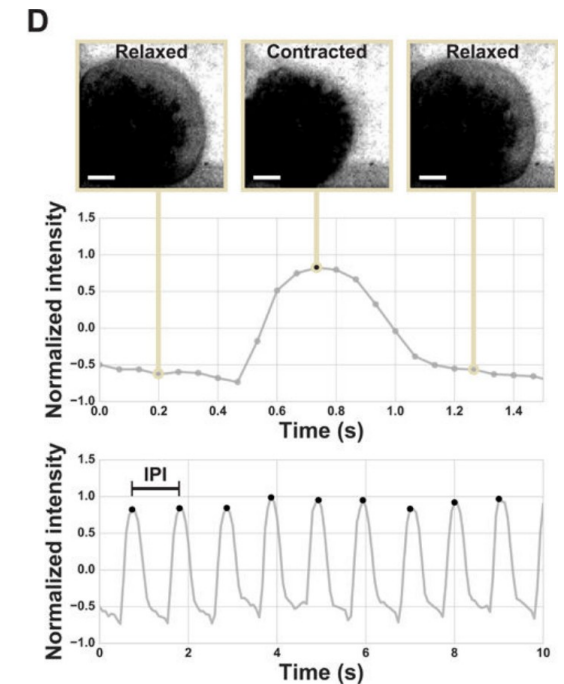
*Cassiopea* is an upside-down **scyphozoan**, a jellyfish belonging to the class Scyphozoa. Its **bell** is the contractile body disc; the oral arms face upward while the bell rests on a surface. Nath and colleagues recorded this stationary pulsing behavior across repeated 12-h light and dark periods.

Bell relaxation and contraction changed the average brightness of an infrared video region. Peaks in the resulting signal supplied pulse times and the **inter-pulse interval**, the elapsed time between successive contractions. Typical ongoing pulsation was approximately one contraction per second, although individual rates varied.

A continuously contracting animal can remain in the same location. This motor rhythm therefore requires measurements of contractions rather than displacement alone. The video signal quantifies behavioral output; it does not identify the neurons that initiate a pulse or directly measure their electrical activity.



Nath et al. (2017), Fig. 1(B,C). Adult *Cassiopea* and labeled rhopalial tissue.



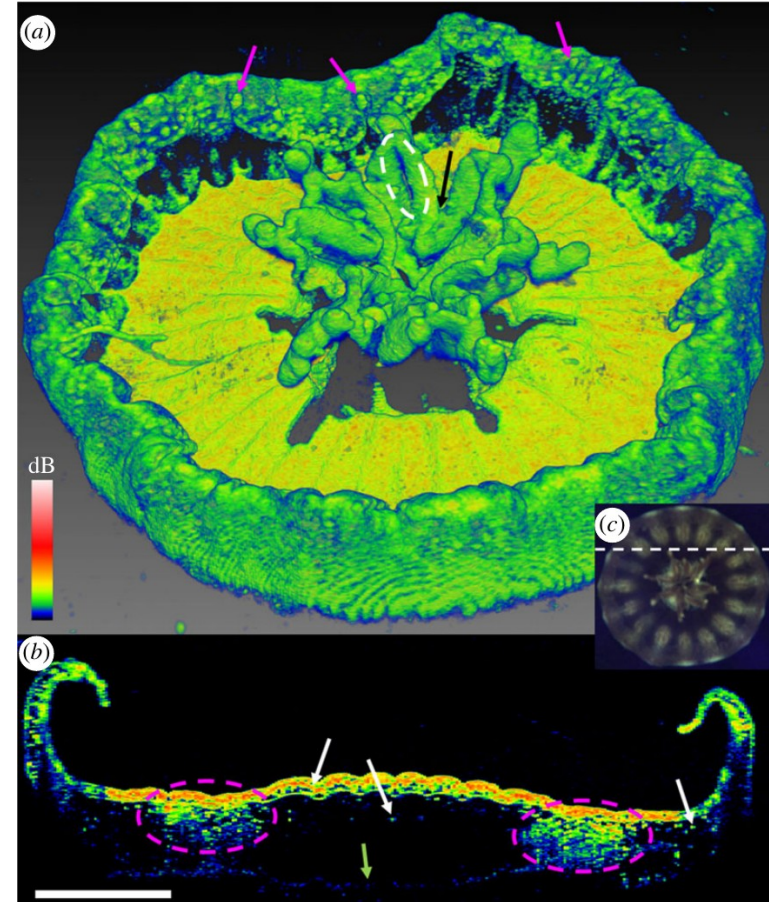
Nath et al. (2017), Fig. 1(D). Bell contractions and the measured inter-pulse interval.

# The bell contains distinct tissue compartments

The **subumbrella** is the oral-facing bell surface; the **exumbrella** is its opposite surface. Between tissue layers lies **mesoglea**, the extracellular material that forms much of the bell. Lyndby and colleagues examined juvenile *Cassiopea* using **optical coherence tomography**, or OCT, which resolves tissue structure through reflected near-infrared light.

A whole-medusa scan covered  $7 \times 7 \times 2.8$  mm. Successive sections separated the bell covering, internal canals and oral structures without requiring the intact animal to be sectioned. Signal strength differed among tissues, and strongly scattering structures were distinguishable from surrounding material.

These measurements establish spatial relationships among bell compartments. OCT brightness represents optical scattering, rather than neuronal firing or a neuron-specific label. Consequently, a canal or tissue boundary can be localized by OCT, while identifying the associated nerve cells requires additional anatomical markers.



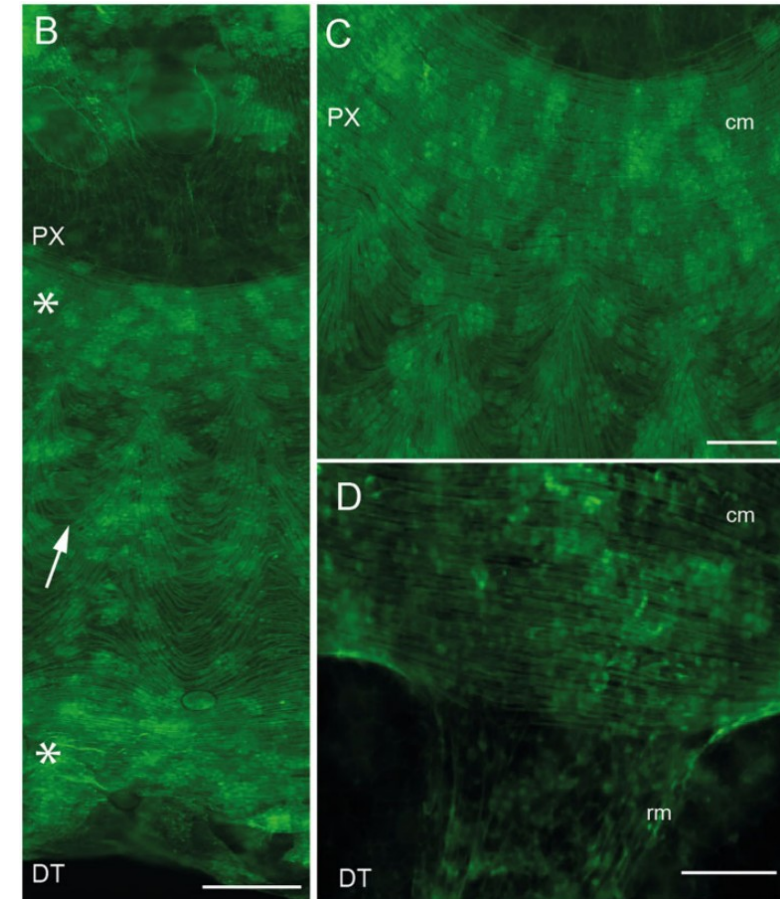
Lyndby et al. (2023), Fig. 1(a–c). Original OCT volume and section through the juvenile bell.

# Circular and radial muscle fibers occupy different regions

An **ephyra** is the young medusa released from a scyphozoan polyp. Muffett and colleagues mapped the bell musculature of *Cassiopea xamachana* ephyrae with **phalloidin**, a probe that binds filamentous actin, the protein polymer enriched in muscle contractile structures. Their anatomical preparations included ephyrae approximately 5 mm in diameter.

A broad **coronal muscle band** surrounded the oral-arm bases and extended toward the bell rim. Closely arranged circular fibers occurred at its inner and outer edges. Between these edges, fibers followed an undulating pattern; within the **lappets**, marginal bell lobes, the musculature became radially oriented.

The spatial arrangement supplies an anatomical basis for coordinated bell deformation. Actin labeling identifies fiber organization, but it does not record which fibers become active first or establish their synaptic inputs. Muscle geometry and neural control are separate measured properties.



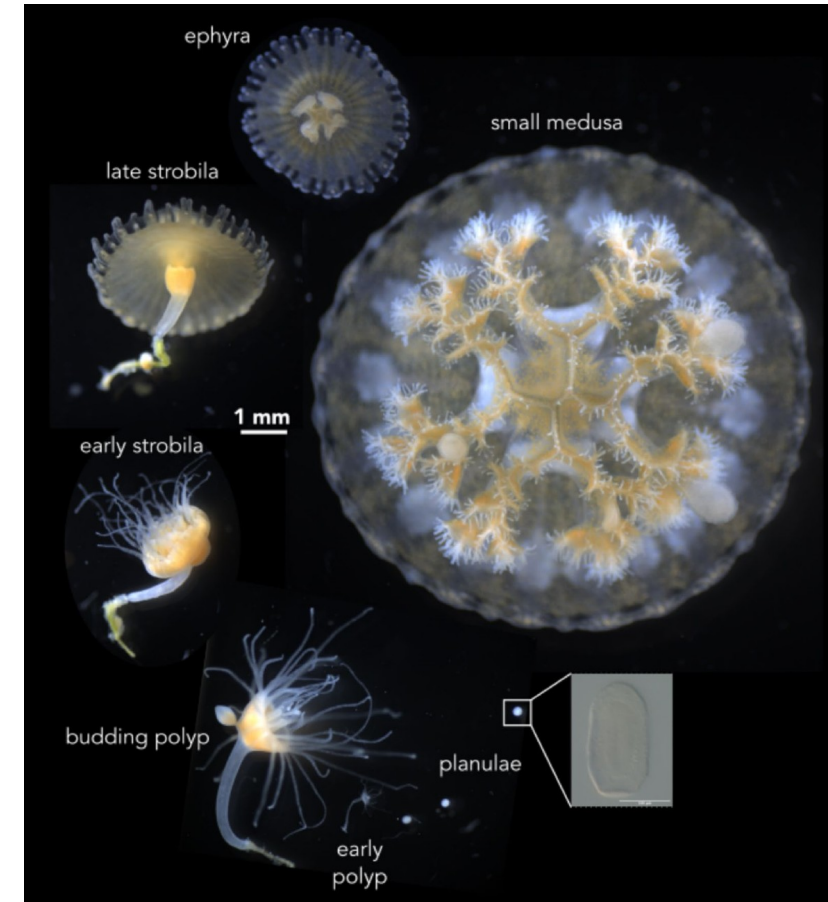
Muffett et al. (2025), Fig. 5(B–D). Actin-stained coronal muscle band and radial fibers.

# Different life stages have different motor structures

*Cassiopea xamachana* alternates between a sexually reproducing medusa and a sessile **polyp**, a tentacle-bearing form attached by its aboral end. Fertilized eggs produce a ciliated **planula larva**. **Strobilation** transforms a polyp region into an ephyra, whereas a **planuloid bud** is an asexually produced swimming propagule.

Steinworth and Martindale documented these stages at common photographic scale, with a 1-mm reference and a separate larval inset. Amplatz and colleagues found that settlement-to-polyp metamorphosis took about 3 days at 20–22°C under their culture conditions; detached planuloid buds developed into polyps over approximately 3–6 days.

A larval ciliary system, a polyp neuromuscular system and a medusa bell-control system are different preparations. Results from one stage cannot automatically identify the circuit of another. Developmental anatomy establishes when structures appear and how their positions change.



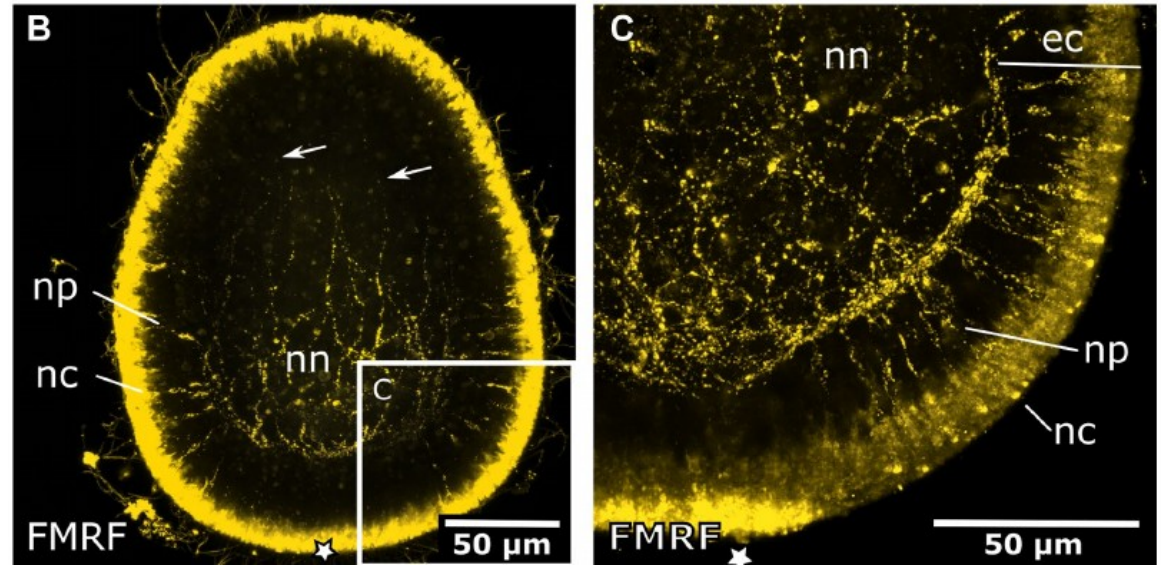
Steinworth & Martindale (2025), Fig. 1. Photographed life-history stages, including the swimming ephyra and medusa.

# Larval neurites form a basiepithelial plexus

A **neurite** is a neuronal process that carries signals away from or toward the cell body. In *Cassiopea* planulae, Amplatz and colleagues used antibody labeling and **confocal microscopy**, which collects optical sections through fluorescent specimens, to localize neurites and candidate sensory cells.

Antibodies against FMRFamide produced **FMRFamide-like immunoreactivity**: binding consistent with a related peptide antigen, rather than chemical identification of an exact peptide. Labeled processes formed a **basiepithelial plexus**, a neural meshwork near the base of the surface tissue. Longitudinal bundles extended across more than three quarters of the larval length.

Primary- and secondary-antibody omission controls tested background binding. The distribution establishes a spatially organized larval network, but antibody labeling does not measure peptide release, synaptic strength or spike propagation. The exact transmitter identity remains distinct from the observed immunoreactivity.



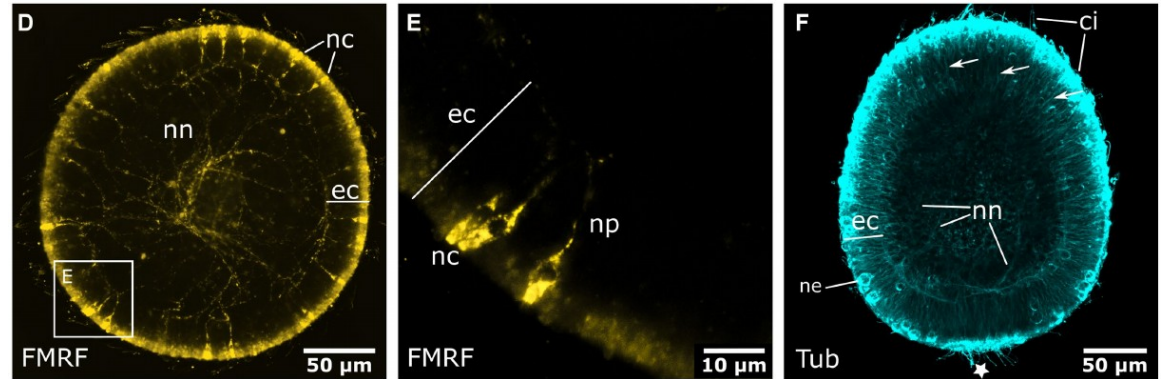
Amplatz et al. (2024), Fig. 4(B,C). FMRFamide-like labeling of the larval basiepithelial neural plexus.

# Larval sensory-cell bodies are spatially polarized

A **sensory neuron** receives information through a specialized cell region and relays it through neuronal processes. In early *Cassiopea* planulae, FMRFamide-like cell bodies occupied the superficial **ectoderm**, the outer embryonic tissue layer, predominantly in the aboral half. Their processes extended inward toward the basal neural meshwork.

Amplatz and colleagues compared peptide-like labeling with tyrosinated  $\alpha$ -tubulin labeling. **Tubulin** is a structural component of microtubules and also occurs in cilia and other nonneuronal structures. The larval preparations contained both basal labeled neurites and thin tubulin-positive fibers throughout the ectoderm.

Omission controls revealed nonspecific secondary-antibody binding near the aboral tip, so that region could not establish the presence or absence of an apical neural organ. The supported result is a polarized distribution outside that ambiguous region, rather than a complete census of every larval sensory cell.



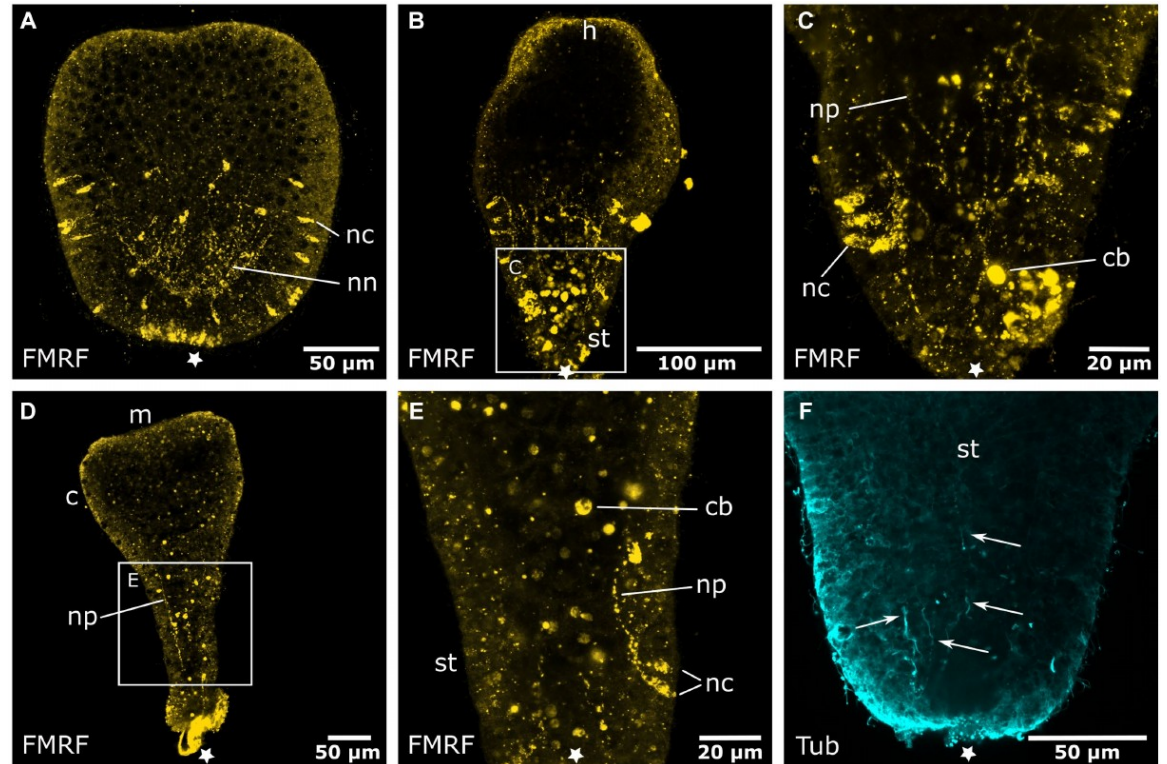
Amplatz et al. (2024), Fig. 4(D–F). Sensory-cell distribution and tubulin-like labeling in planulae.

# Settlement remodels the larval nervous system

During **metamorphosis**, a larva changes its body organization as it becomes a polyp. Amplatz and colleagues followed *Cassiopea* settlement through a hood-shaped stage, mouth formation and stalk elongation, using the same peptide-like and tubulin-like neural markers across developmental stages.

Long FMRFamide-like neurites in the oral half diminished first. Labeled cell bodies subsequently became less numerous and irregularly shaped, and scattered labeled fragments remained during stalk formation. The aborally concentrated larval labeling was no longer detectable by the tentacle stage; polyp metamorphosis required approximately 3 days at 20–22°C.

The changing distribution supports extensive neural reorganization rather than simple enlargement of an unchanged larval net. **Apoptosis**, regulated cell death, was proposed as an explanation for disappearing cells, but a specific cell-death assay was not performed. Loss of immunoreactivity alone cannot distinguish cell death from altered marker expression.



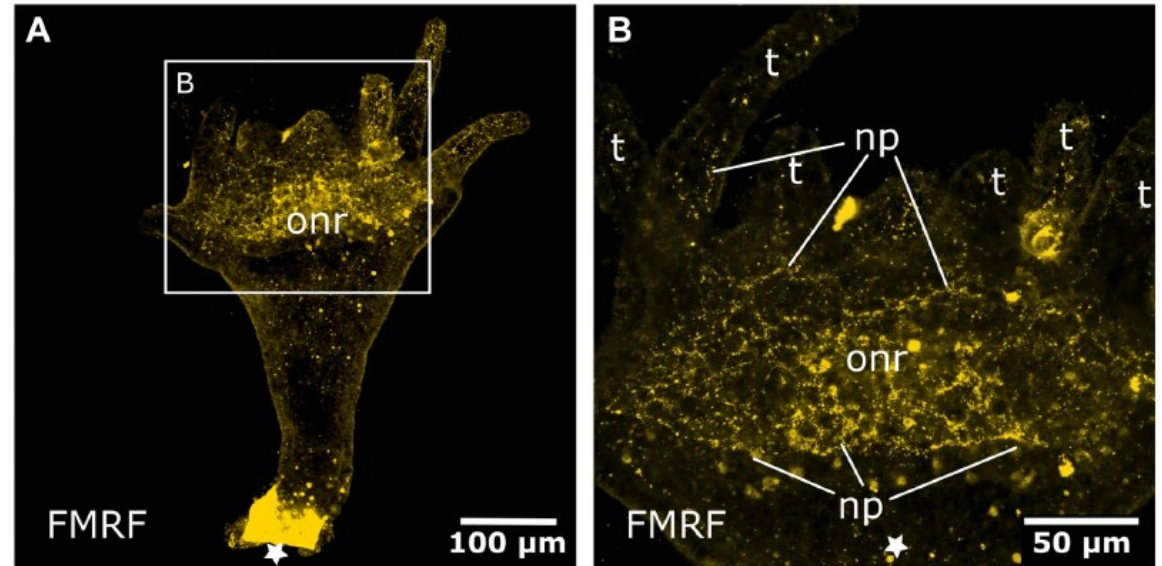
Amplatz et al. (2024), Fig. 5(A–F). Reduction and remodeling of larval neural labeling during settlement.

# Polyp neurites concentrate around the oral disc

The **hypostome** is the raised oral region bearing the polyp mouth. Amplatz and colleagues examined *Cassiopea* polyps after tentacle formation and found a new concentration of FMRFamide-like neurites around the **oral disc**, the surface from which the tentacles arise.

In late tentacle stages, labeled processes spanned the oral disc and hypostome. The densest concentration formed a loose **oral nerve ring**, a circumferential arrangement below the tentacle bases. Continued polyp growth elaborated the network, rather than preserving the earlier aboral larval concentration.

The same labeling protocol across stages links the spatial change to development. A ring-shaped anatomical concentration does not establish a single central controller or a direction of information flow. In these preparations, peptide-like neurites were clear, while corresponding immunoreactive cell bodies were not consistently identifiable.



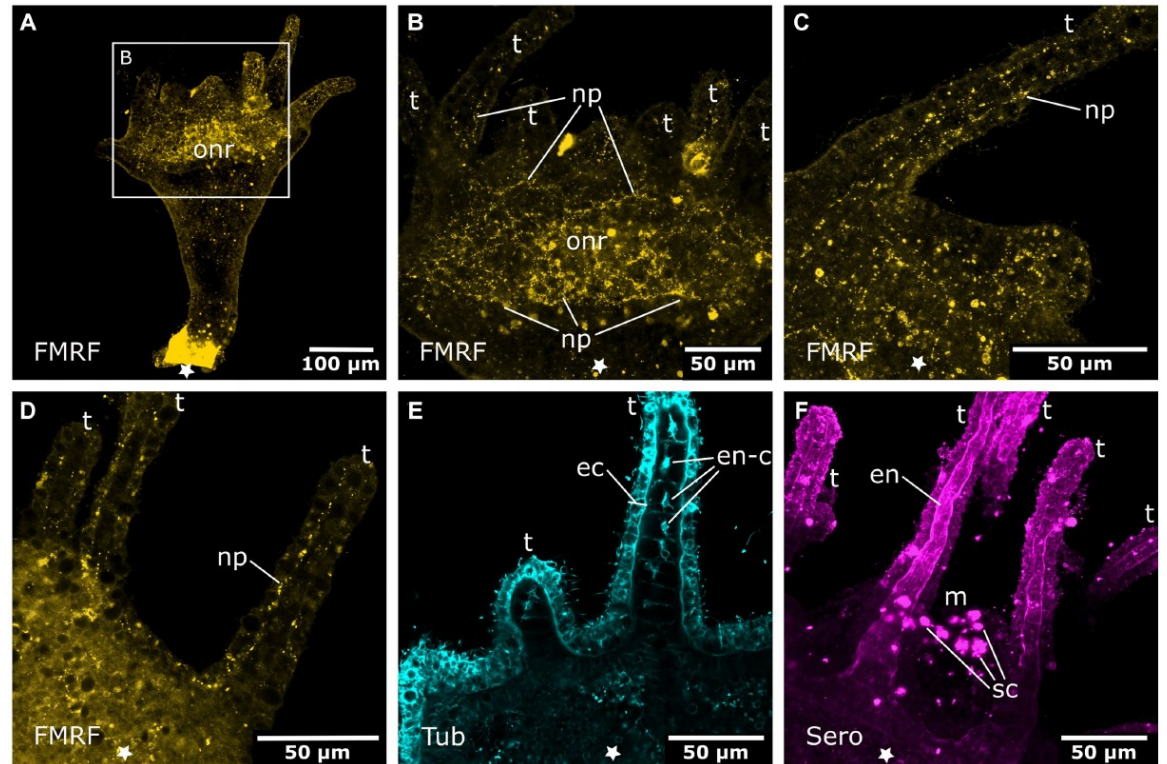
Amplatz et al. (2024), Fig. 7(A,B). Oral FMRFamide-like neurites in developing polyps.

# Polyp tentacles contain branching ectodermal neurites

Polyp tentacles extend the oral disc into the surrounding water. In developing *Cassiopea* polyps, Amplatz and colleagues localized FMRFamide-like neural processes within the tentacle ectoderm using optical sections through the tentacle wall, rather than treating the entire tentacle as a single structure.

The labeled processes branched around the ectoderm and were continuous with the broader oral-region network. The adjacent **endoderm** contained large, vacuolated discoidal cells arranged in a single axial row. These inner cells differed in position and morphology from the fine neural processes in the outer tissue.

Comparing tissue layers prevents assignment of every tentacle signal to the same cell class. The branching ectodermal network establishes anatomical innervation, but the observations do not identify a specific sensory stimulus, transmitter receptor or muscle response generated by each branch.



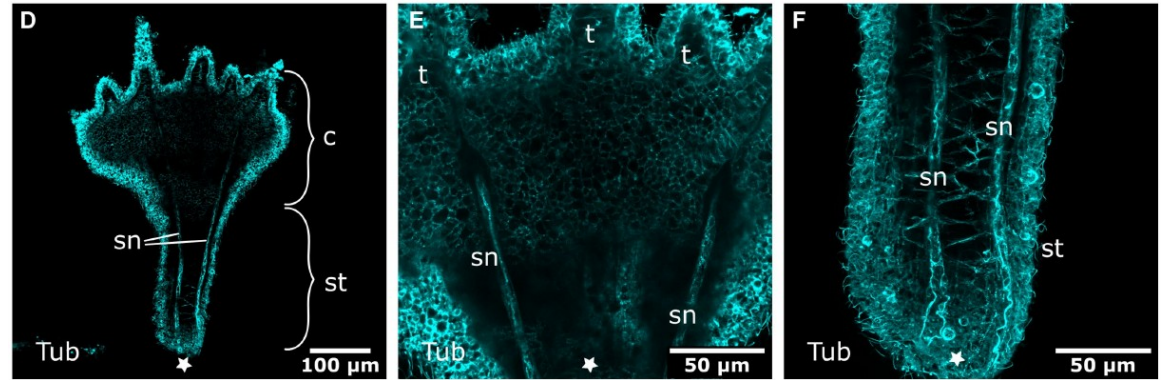
Amplatz et al. (2024), Fig. 7(A–F). Oral and tentacle neural labeling in developing polyps.

# Septal neurites accompany four muscle bands

The polyp **stalk** links the expanded **calyx**, or body cup, to the attachment region. Amplatz and colleagues detected four prominent **septal muscle bands**, longitudinal structures extending through this body axis, and compared actin labeling with tubulin-like labeling in developing *Cassiopea* polyps.

Distinct tubulin-like **septal neurites** ran alongside the four bands from the foot and stalk toward the tentacle bases. Each band was accompanied by several processes with branches. Additional thin longitudinal processes were distributed across the outer stalk and calyx.

Spatial correspondence supports a differentiated neuromuscular arrangement within the polyp. It does not establish that every adjacent neurite directly synapses onto the neighboring muscle or that one neuron controls an entire band. The comparison with actin staining separates candidate neural fibers from contractile structures.



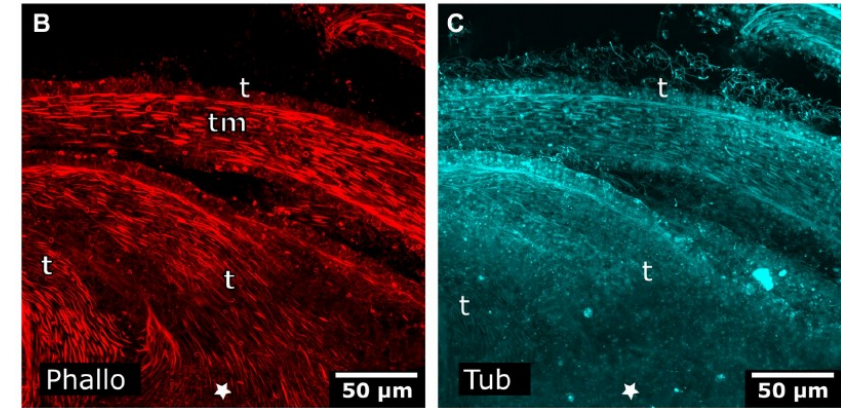
Amplatz et al. (2024), Fig. 9(D–F). Tubulin-like septal neurites alongside the four septal muscle bands.

# Tubulin labeling can include nonneuronal structures

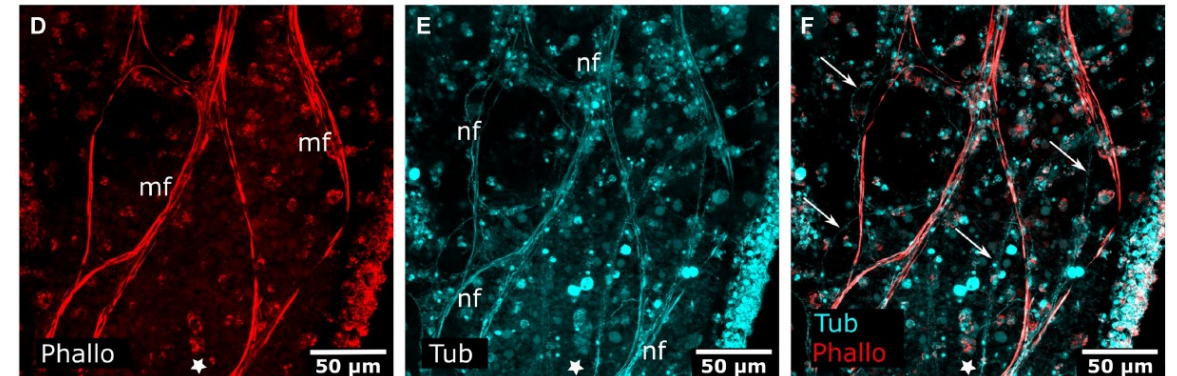
A **cell marker** is a molecular feature used to recognize a cell or structure. Its usefulness depends on specificity. Amplatz and colleagues compared phalloidin-labeled actin with tyrosinated tubulin-like labeling in the same *Cassiopea* polyp regions, including tentacles and developing bud sites.

Tentacle tubulin labeling closely resembled muscle labeling and also included **nematocytes**, stinging cells, and cilia. At bud-forming sites, some fine tubulin-positive fibers lacked the corresponding phalloidin signal and could be distinguished from muscles. Antibody-omission controls also identified intrinsic background and secondary-antibody binding.

Tubulin fluorescence alone therefore cannot define a complete neuronal population. Combining tissue location, cell morphology, actin comparison and omission controls narrows the interpretation. Absence of a clearly separable neural signal in the tentacle does not establish absence of nerves there.



Amplatz et al. (2024), Fig. 8(B,C). Same polyp tentacle regions labeled for actin and tubulin.



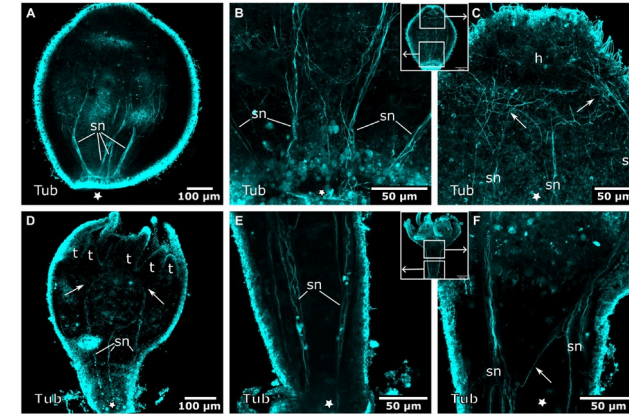
Amplatz et al. (2024), Fig. 8(D–F). Bud-forming tissue with muscle and neuronal-fiber labeling.

# Asexual buds inherit a different neural starting pattern

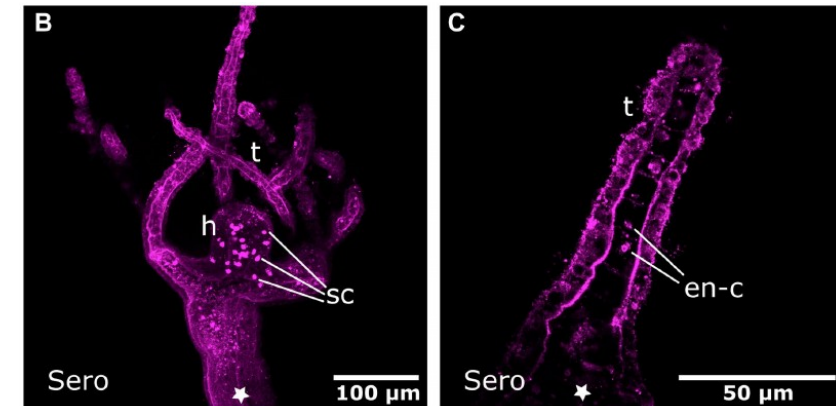
A *Cassiopea* **planuloid bud** forms from an adult polyp and separates as a swimming propagule. Amplatz and colleagues followed its neural development with tubulin-like and serotonin-like antibody labeling, then compared the stages with those of sexually produced planula larvae.

Early detached buds already contained four septal muscle strands and associated tubulin-like neuronal strands. These processes extended orally, branched and later connected near the stalk–calyx boundary. By the young-polyp stage, the principal neural arrangements resembled those of planula-derived polyps; detached-bud development took approximately 3–6 days at 20–22°C.

The two routes reach similar polyp anatomy from different starting tissue organizations. **Serotonin-like immunoreactivity** also occurred in endodermal cells without visible neurites, so it did not establish that every labeled cell was neuronal or released serotonin as a transmitter.



Amplatz et al. (2024), Fig. 10(A–F). Septal neuronal strands during asexual planuloid development.



Amplatz et al. (2024), Fig. 9(B,C). Serotonin-like labeling in hypostome and discoidal tentacle endoderm.

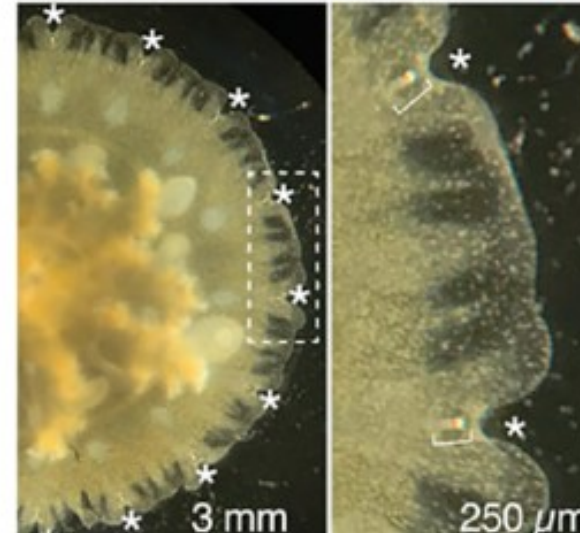
# Rhopalia are distributed around the medusa margin

A **rhopalium** is a marginal sensory structure containing neural tissue associated with bell-rhythm control. In *Cassiopea* medusae, multiple rhopalia are radially spaced around the bell rim. Their distribution combines a widespread control architecture with localized concentrations of neurons.

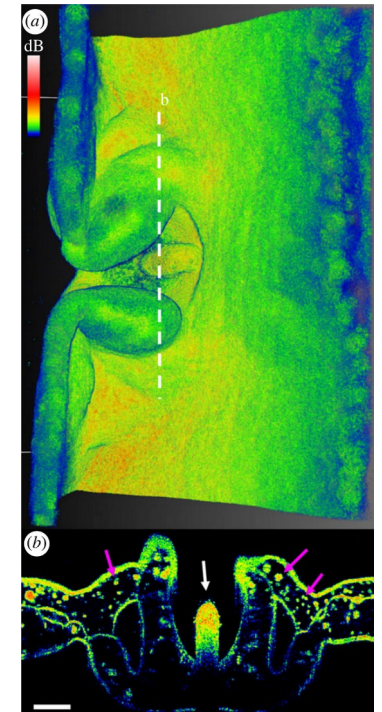
Abrams and colleagues examined the marginal organs directly and sampled rhopalial tissue separately from the remaining bell. Lyndby and colleagues resolved rhopalia in intact medusae approximately 1 and 4 cm in diameter with OCT. A canal extended to each organ's base and into the neighboring marginal tissue.

These observations identify repeated sensory–neural sites rather than one head-localized brain. The presence of several organs does not mean they fire independently during every bell pulse. Anatomical distribution establishes their locations; coordination and rhythm generation require functional tests.

## C Radial rhopalia (ganglia)



Abrams et al. (2025), Fig. 1(C). Radial distribution of rhopalia and a higher-magnification rhopalium.



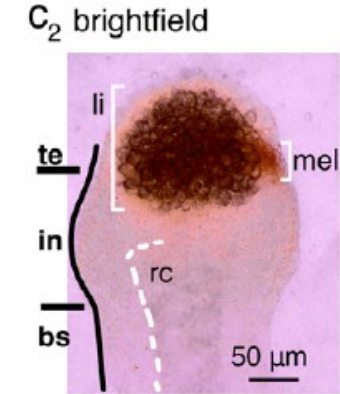
Lyndby et al. (2023), Fig. 2(A,B). Surface and section through the rhopalium and its associated canal.

# A rhopalium contains three anatomical segments

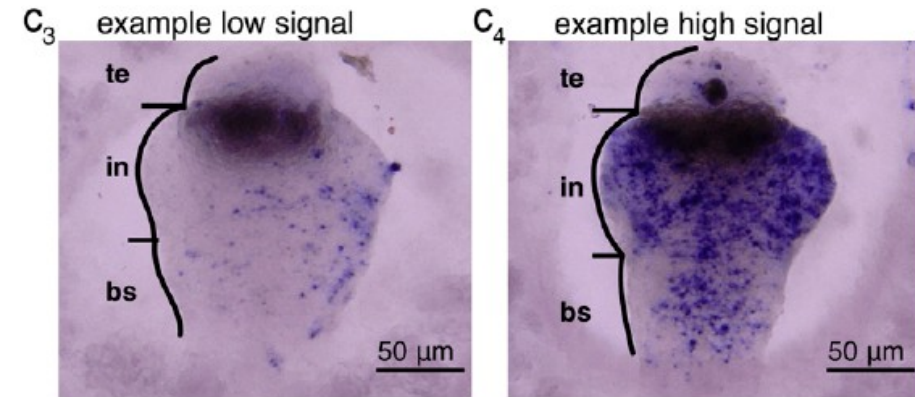
Abrams and colleagues divided the *Cassiopea* rhopalium into **terminal**, **intermediate** and **basal** segments. The terminal end projects away from the bell; the basal region connects the organ to the medusa. Their bright-field and gene-expression preparations preserved the relation among these regions.

A dark **lithocyst**, the statolith-containing balance structure, occupied the terminal region. Pigmented tissue associated with a **spot ocellus**, a simple light-sensitive organ, lay near the terminal–intermediate boundary. The **radial canal** approached the organ from its basal side, while *chrnal-E* expression occupied rhopalial neural regions.

The 50- $\mu$ m reference scale permits comparison of localized tissue compartments without treating the whole organ as one cell. Gross segment boundaries do not locate every pacemaker neuron. The published ganglion-group diagram was adapted from *Aurelia*, so that detailed comparative map is not a complete *Cassiopea* cell census.



Abrams et al. (2025), Fig. 3(C,c2). *Cassiopea* rhopalium with terminal, intermediate and basal regions.



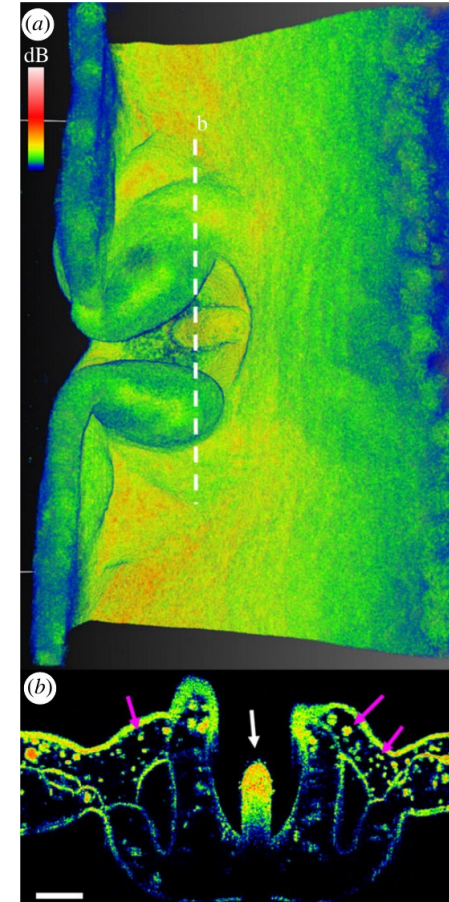
Abrams et al. (2025), Fig. 3(C,c3,c4). Low and high *chrnal-E* expression in rhopalial tissue.

# Rhopalial canals connect organs with bell tissues

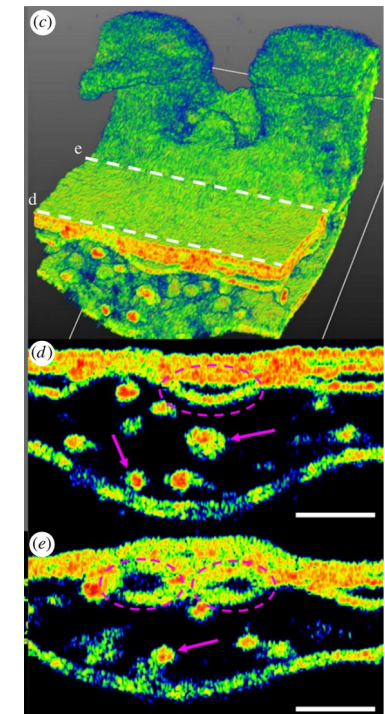
Lyndby and colleagues used noninvasive OCT to examine *Cassiopea* rhopalia and their neighboring bell tissue. Serial sections localized the rhopalial canal, its branches and the strongly scattering sensory organ while preserving the organ's attachment to the bell margin.

In medusae approximately 1 cm in diameter, the canal extended to the rhopalial base and split toward marginal tissues. Sections also distinguished dense scattering granules from clusters of algal symbionts. **Symbionts** are organisms living in a close biological association with a host; these intracellular algae support *Cassiopea* photosymbiosis.

The measured anatomy places sensory and neural structures within a canal-connected, optically heterogeneous margin. OCT does not identify a synapse or distinguish sensory from pacemaker cells by itself. Strong scattering is a tissue property, rather than evidence that the rhopalium is electrically active at that moment.



Lyndby et al. (2023), Fig. 2(A,B). Surface and section through the rhopalium and its associated canal.



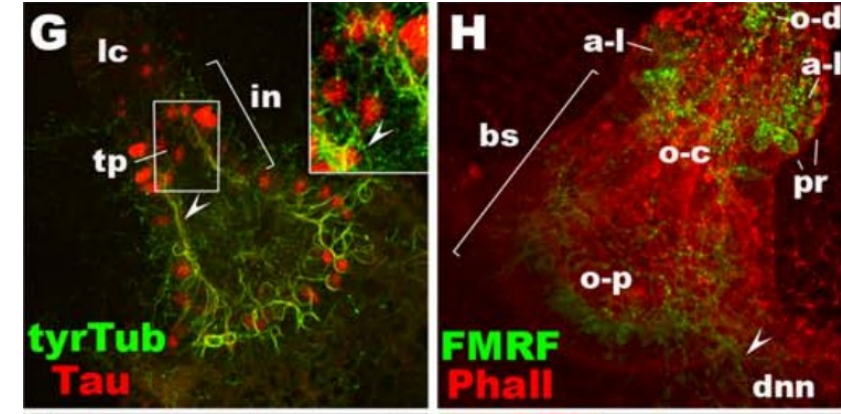
Lyndby et al. (2023), Fig. 2(c–e). Rhopalial canal and adjacent bell tissues in original OCT sections.

# Comparative rhopalia contain mechanosensory cells

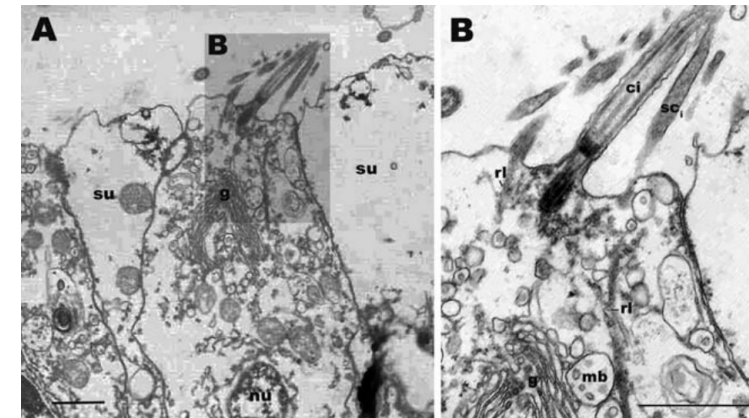
Nakanishi and colleagues examined rhopalia of *Aurelia* sp.1, another scyphozoan, with confocal and electron microscopy. Its **touch plate** is a sensory region associated with the lithocyst. **Mechanosensation** detects physical displacement or force; these structures are candidates for sensing orientation through mechanical loading.

Touch-plate cells had apical cilia and actin-rich rings consistent with associated **stereocilia**, microvillar projections containing actin. Electron microscopy also resolved stereocilia in oral rhopalial cells. Neuronal processes from the touch-plate region joined a tubulin- and taurine-reactive network extending toward the motor net.

The measurements identify sensory specializations and candidate pathways in *Aurelia*. They do not measure force-gated currents or identify the responsible mechanotransduction channel. Similar organs in *Cassiopea* justify a comparative anatomical question, while the detailed ultrastructural findings remain species-specific.



Nakanishi et al. (2009), Fig. 3(G,H). Touch-plate fibers and FMRFamide-like rhopalial projections.



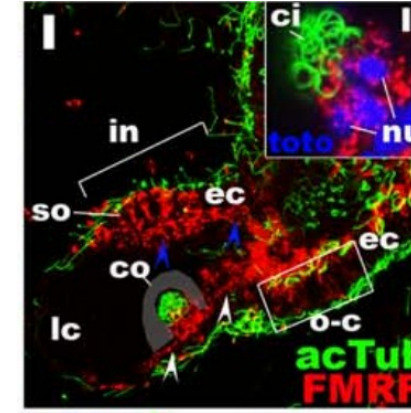
Nakanishi et al. (2009), Fig. 4(A,B). Electron micrographs of oral rhopalial cilia and stereocilia in *Aurelia*.

# Comparative ocelli connect to rhopalial neurites

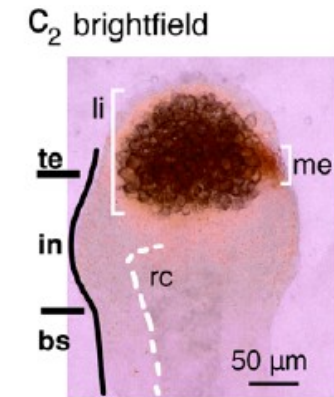
An **ocellus** is a simple photoreceptive structure. Nakanishi and colleagues identified a pigment-cup ocellus on the oral side and a pigment-spot ocellus on the aboral side of *Aurelia* rhopalia, using optical sections that separated pigment-bearing layers from labeled sensory processes.

The cup organ contained ciliated sensory cells facing a pigment-cell layer. Their basal processes joined the FMRFamide-reactive rhopalial network. In the spot organ, labeled pigmented cells also extended processes toward that network. Cassiopea bright-field anatomy independently localized pigmented spot-ocellus tissue near its terminal rhopalial region.

Anatomy supports sensory access to local neural networks. It does not establish visual acuity, an image-forming retina or the sign of a light response. The *Aurelia* cellular connections are comparative evidence; the *Cassiopea* pigment location alone cannot establish identical cellular wiring.



Nakanishi et al. (2009), Fig. 3(I). Comparative cup and spot ocelli with sensory-cell processes.



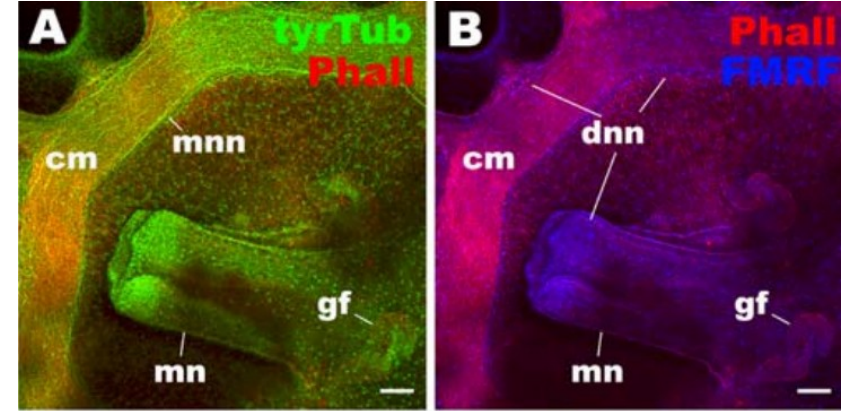
Abrams et al. (2025), Fig. 3(C,c2). Cassiopea rhopalium with terminal, intermediate and basal regions.

# Two comparative nerve nets have distinct distributions

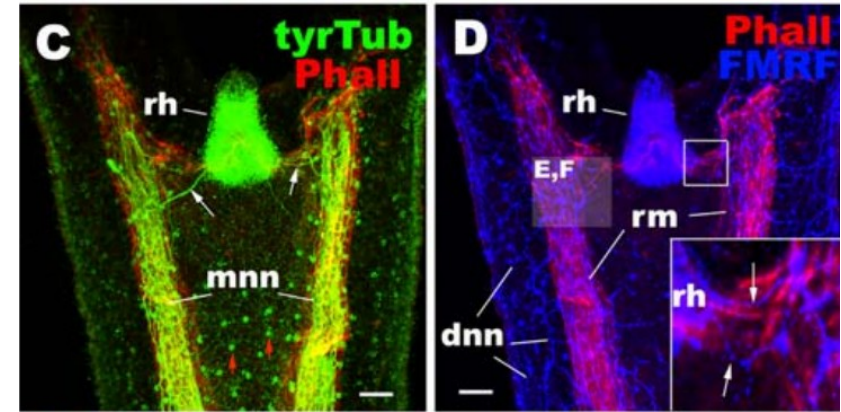
A **nerve net** is an interconnected arrangement of neuronal processes distributed through tissue. Nakanishi and colleagues distinguished a **motor nerve net**, or MNN, and a **diffuse nerve net**, or DNN, in *Aurelia* ephyrae by combining antibody labeling with actin-labeled muscle anatomy.

Thick, strongly tubulin-reactive motor-net processes spread through the subumbrella and connected with rhopalia. FMRFamide-reactive diffuse-net fibers were finer and associated with both subumbrellar and exumbrellar surfaces. In the same specimens, the two labeling patterns followed different paths near the rhopalial base.

Multiple spatial patterns contradict a description of a nerve net as one anatomically uniform sheet. Marker-defined pathways constrain possible functions, but anatomy alone cannot measure conduction speed or assign every behavioral response to one net. This detailed separation was measured in *Aurelia*, rather than directly mapped throughout adult *Cassiopea*.



Nakanishi et al. (2009), Fig. 2(A,B). Comparative *Aurelia* motor and diffuse nerve-net labeling.



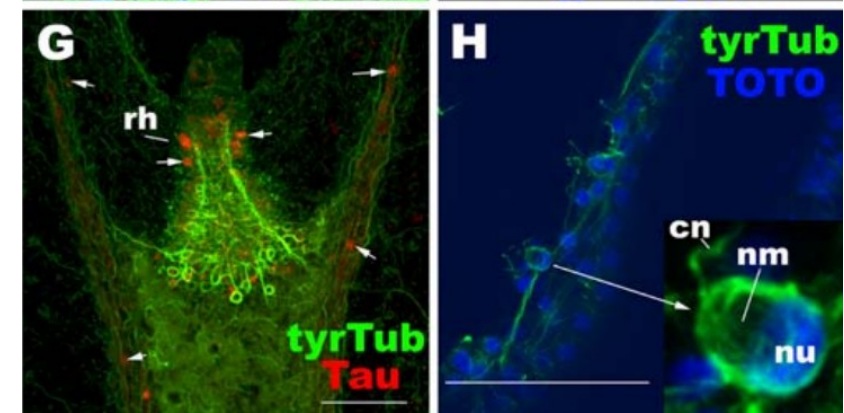
Nakanishi et al. (2009), Fig. 2(C,D). Distinct rhopalial connections of the two *Aurelia* nerve nets.

# Motor-net processes extend between radial regions

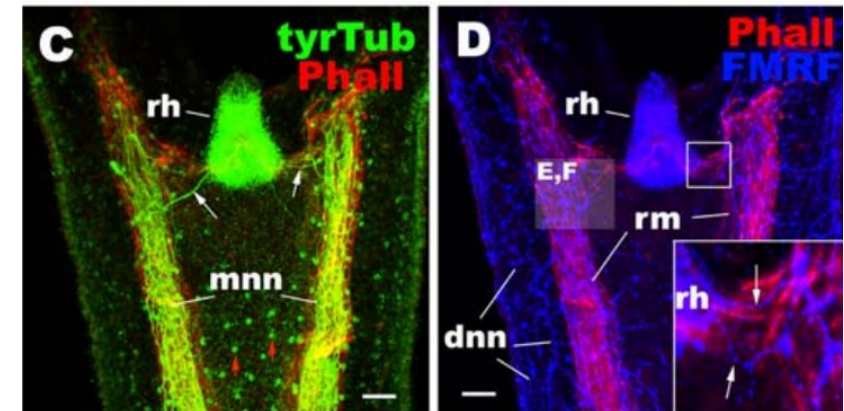
The comparative *Aurelia* motor net contains thick neuronal processes spanning the subumbrellar surface between rhopalia and muscle-bearing bell regions. Nakanishi and colleagues labeled these processes with tyrosinated tubulin and taurine antibodies while phalloidin identified the surrounding muscle fibers.

The labeled motor-net neurons were predominantly **bipolar**, with two principal processes extending from the cell body. Thick projections from the rhopalial base continued into this net, providing an anatomical route between local sensory–neural tissue and the distributed bell musculature.

**Connectivity** means a physical or functional relation between circuit components, but light-microscopic continuity is not a complete synaptic reconstruction. A process crossing another process does not establish a synapse, the specialized junction through which cells communicate. The source constrains circuit geometry without resolving every connection or its direction.



Nakanishi et al. (2009), Fig. 2(G,H). Comparative bipolar motor-net processes and cnidocyte structure.



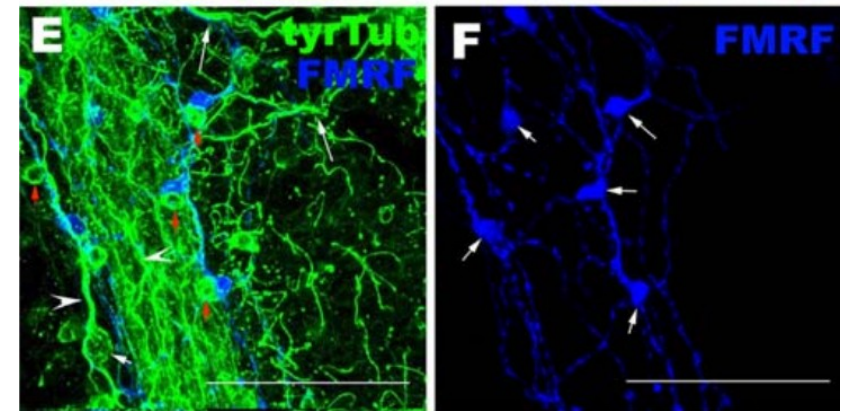
Nakanishi et al. (2009), Fig. 2(C,D). Distinct rhopalial connections of the two *Aurelia* nerve nets.

# Diffuse-net neurons have branching processes

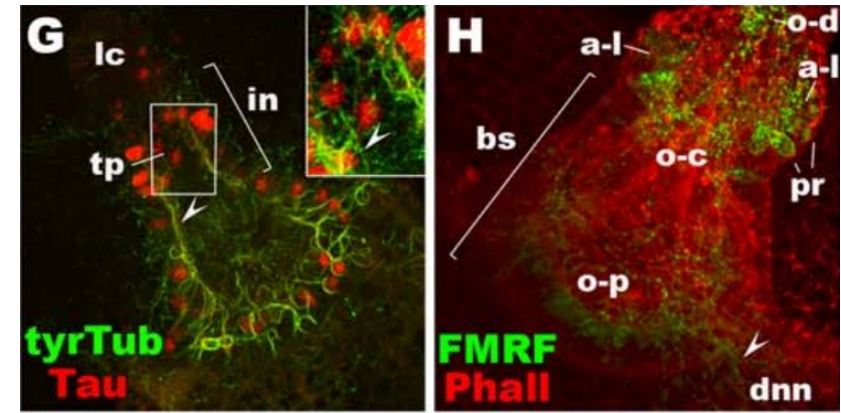
In *Aurelia* ephyrae, FMRFamide-reactive diffuse-net neurons were largely **multipolar**, with several branching processes instead of the two prominent extensions typical of the motor-net cells. Nakanishi and colleagues resolved their morphology within the same bell and rhopalial preparations.

Fine labeled processes lay near **cnidocytes**, stinging cells with a discharge apparatus, and extended from rhopalial tissue toward the surrounding net. Their distribution differed from that of the thicker motor-net processes. Cells and tissue depth were distinguished with tubulin labeling and nuclear markers.

Close association supports candidate sensory and effector pathways, but it does not establish direct control of cnidocyte discharge in this experiment. Neither peptide identity nor signaling direction follows from immunoreactivity alone. The contrasting morphologies establish differentiated network components in the comparative scyphozoan preparation.



Nakanishi et al. (2009), Fig. 2(E,F). Comparative multipolar diffuse-net neurons and nearby cnidocytes.



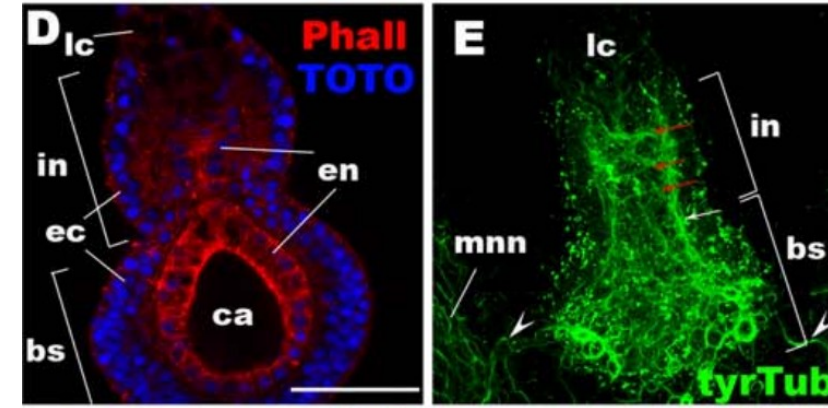
Nakanishi et al. (2009), Fig. 3(G,H). Touch-plate fibers and FMRFamide-like rhopalial projections.

# Rhopalial processes join different peripheral networks

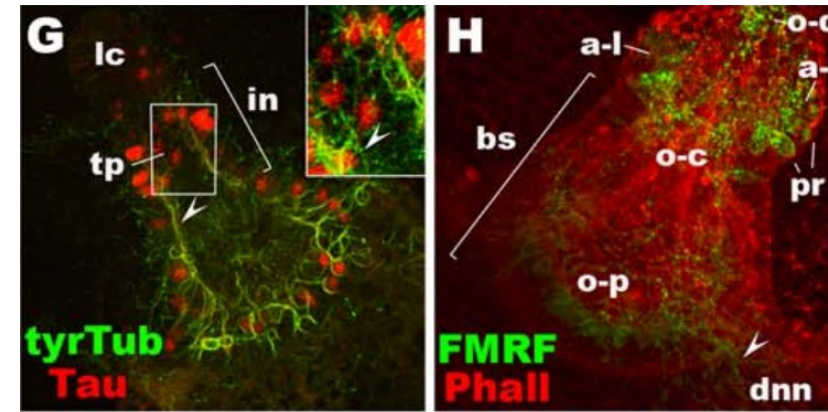
Nakanishi and colleagues traced marker-defined neuronal processes through *Aurelia* rhopalia using confocal sections. The basal segment contained dense ectodermal neural tissue around a canal; neural processes extended longitudinally through the intermediate region and crossed between its two sides.

Tubulin- and taurine-reactive fibers left the basal region toward the motor net. FMRFamide-reactive fibers preferentially left lateral basal regions toward the diffuse net. Other GLWamide-reactive cells, recognized by an antibody against a different neuropeptide family, had processes confined to the rhopalium.

Distinct exits and internal distributions support differentiated sensory–neural organization within each organ. They do not establish the exact cells that generate the pacemaker rhythm. Local neuropeptide immunoreactivity also cannot specify whether a given connection excites, inhibits or modulates its target.



Nakanishi et al. (2009), Fig. 3(D,E). Rhopalial ectoderm and motor-net projections in *Aurelia*.



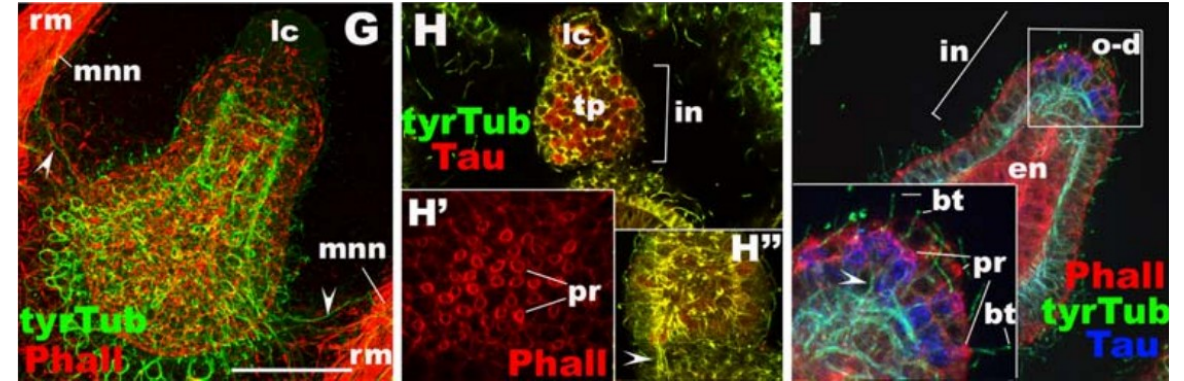
Nakanishi et al. (2009), Fig. 3(G,H). Touch-plate fibers and FMRFamide-like rhopalial projections.

# Neural development precedes coordinated beating

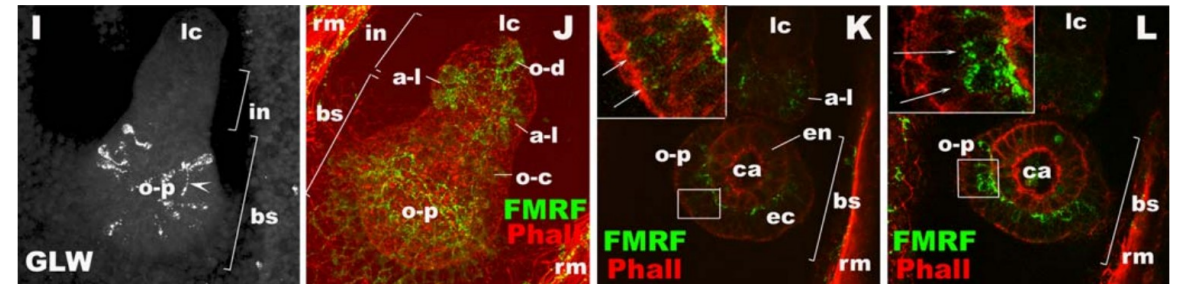
Nakanishi and colleagues examined successive *Aurelia* developmental stages while the future ephyrae remained attached to the strobilating polyp. A **prephyra** is one of these developing medusa units before liberation. Sensory structures, neural markers and beating behavior appeared in an ordered sequence.

Statolith formation began early. By the prephyra III stage, rhopalar arms contracted together spontaneously, and labeled rhopalial processes connected with the motor net. Ocellar differentiation occurred later than the initial mechanosensory structures and the onset of coordinated beating.

The temporal association places motor-network differentiation near emergence of the rhythm, without identifying a single oscillator cell. These comparisons concern fixed developmental stages and observed contractions, rather than recordings from every candidate neuron. Their timing in *Aurelia* cannot be assigned unchanged to *Cassiopea* development.



Nakanishi et al. (2009), Fig. 5(G–I). Mechanosensory and motor-net connections in the beating prephyra.



Nakanishi et al. (2009), Fig. 6(I–L). Later-stage rhopalial neuronal clusters in *Aurelia*.

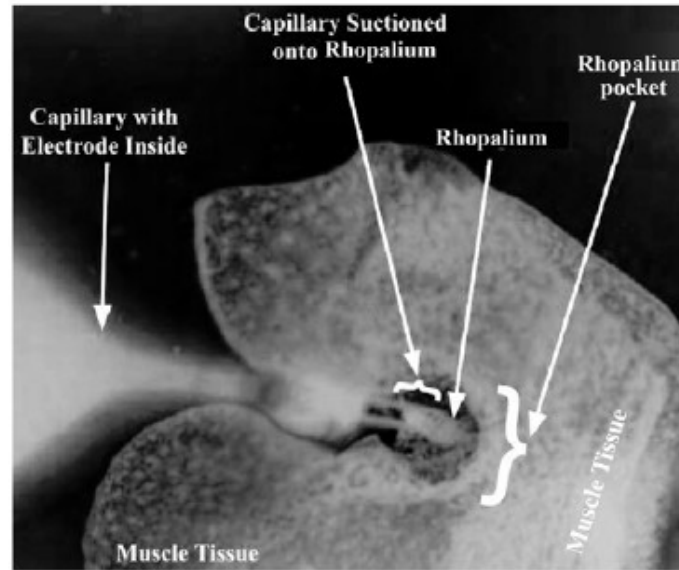
# Isolated rhopalia generate measurable electrical activity

Abrams and colleagues isolated adjacent rhopalia from *Cassiopea* medusae with bell diameters of 3–5 cm. A **suction electrode** holds tissue against a recording tip and detects extracellular electrical signals. The preparation separated activity in these organs from the mechanical movement of the intact bell.

Rhopalia were maintained in filtered artificial seawater at 28°C. Baseline recordings preceded drug application, with electrical signals sampled every 0.08 ms. The **interspike interval**, or ISI, is the elapsed time between detected electrical events and provides a measure of their rhythm.

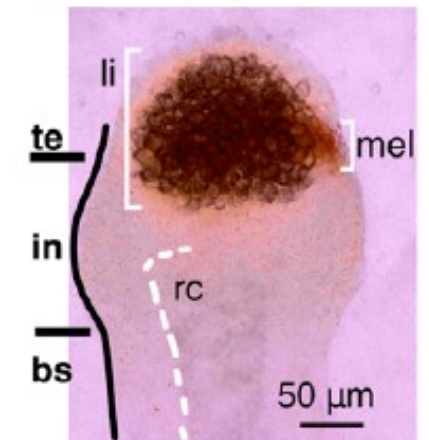
Activity persisting in the isolated preparation localizes a rhythm-generating capability to rhopalial tissue. Extracellular recordings do not identify the membrane potential of one neuron, every participating cell or the exact output reaching the muscle. The organ-level evidence complements, rather than replaces, cell-specific circuit mapping.

b<sub>1</sub> suction electrode setup



Abrams et al. (2025), Fig. 3(B,b1). Original suction-electrode recording preparation.

C<sub>2</sub> brightfield



Abrams et al. (2025), Fig. 3(C,c2). *Cassiopea* rhopalium with terminal, intermediate and basal regions.

# A receptor antagonist slows rhopalial electrical events

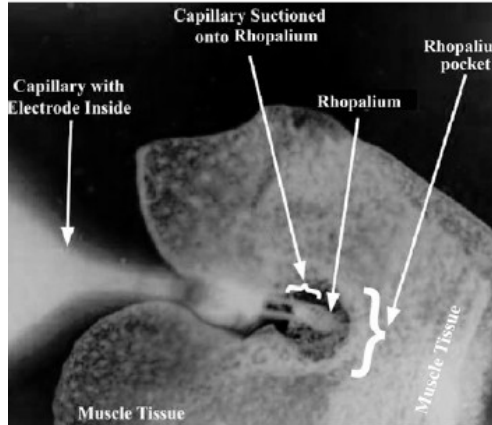
Abrams and colleagues applied **tubocurarine**, a competitive antagonist of nicotinic acetylcholine receptors, to isolated *Cassiopea* rhopalia. An **antagonist** reduces receptor activation; competition means that binding interferes with access of an activating ligand to a receptor-associated site.

The Results and Figure 3 report 500 and 600  $\mu\text{M}$  tubocurarine treatments. Both increased the interspike interval relative to the preceding baseline, with the higher concentration producing a larger slowing. Comparing each preparation before and after treatment controls some differences in baseline rhythmic activity.

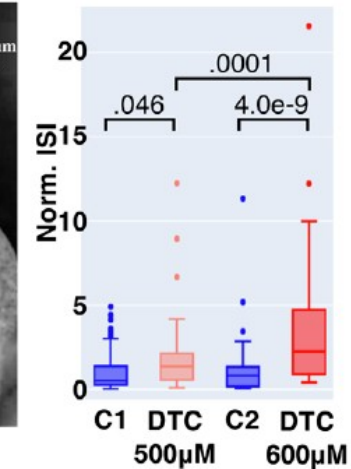
The response supports sensitivity of rhopalial activity to this pharmacological class. It does not identify a single receptor subtype or establish acetylcholine as the endogenous transmitter. Drug action on additional targets remains possible; a receptor-specific genetic perturbation addresses a different level of causation.

## B Pacemaker activity is regulated by cholinergic neuromodulators

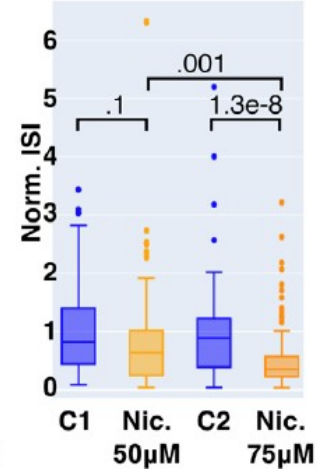
b<sub>1</sub> suction electrode setup



b<sub>2</sub> competitive antagonist: tubocurarine (DTC)



b<sub>3</sub> agonist: nicotine (Nic)



Abrams et al. (2025), Fig. 3(B,b1–b3). Interspike intervals after tubocurarine exposure.

# A receptor agonist accelerates rhopalial electrical events

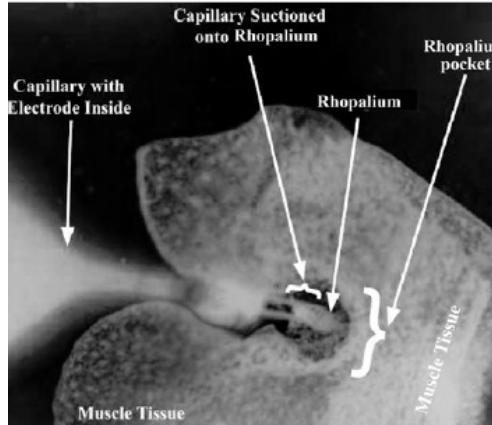
An **agonist** activates a receptor or promotes its activated state. Abrams and colleagues applied nicotine, an agonist of nicotinic acetylcholine receptors in established systems, to the same type of isolated *Cassiopea* rhopalial preparation used for antagonist experiments.

Nicotine at 50 and 75  $\mu\text{M}$  decreased the interspike interval relative to baseline, with the higher concentration producing the larger reduction. The complementary agonist and antagonist responses associate receptor pharmacology with the rate of electrical-event generation in a localized neural organ.

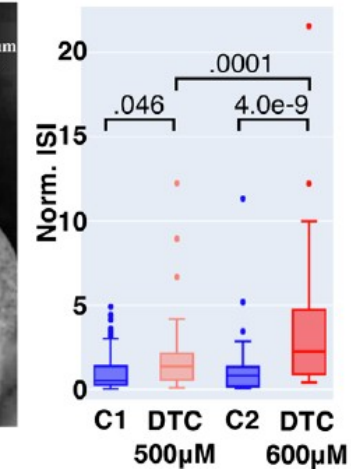
A shorter interval establishes an acceleration under these experimental conditions. It does not demonstrate direct nicotine binding to a particular *Cassiopea* receptor protein. Opposite responses to two drugs strengthen a mechanistic hypothesis while leaving receptor identity, endogenous ligand and individual pacemaker-cell contributions unresolved.

## B Pacemaker activity is regulated by cholinergic neuromodulators

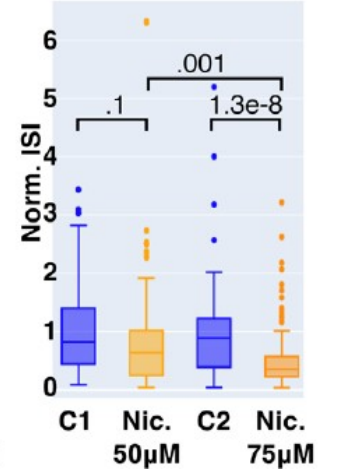
b<sub>1</sub> suction electrode setup



b<sub>2</sub> competitive antagonist: tubocurarine (DTC)



b<sub>3</sub> agonist: nicotine (Nic)



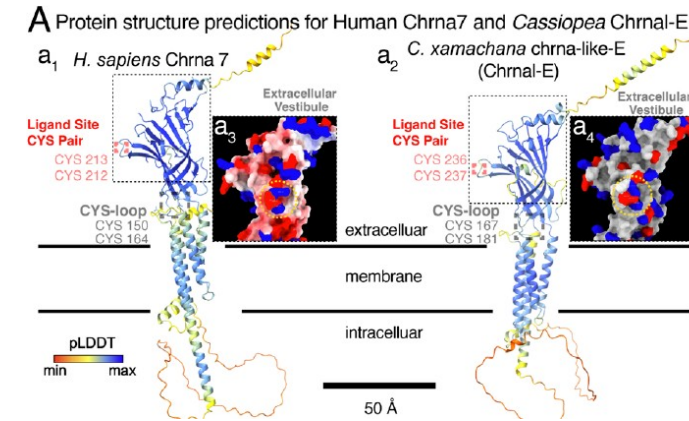
Abrams et al. (2025), Fig. 3(B,b1–b3). Interspike intervals after nicotine exposure.

# A candidate receptor resembles a ligand-gated channel

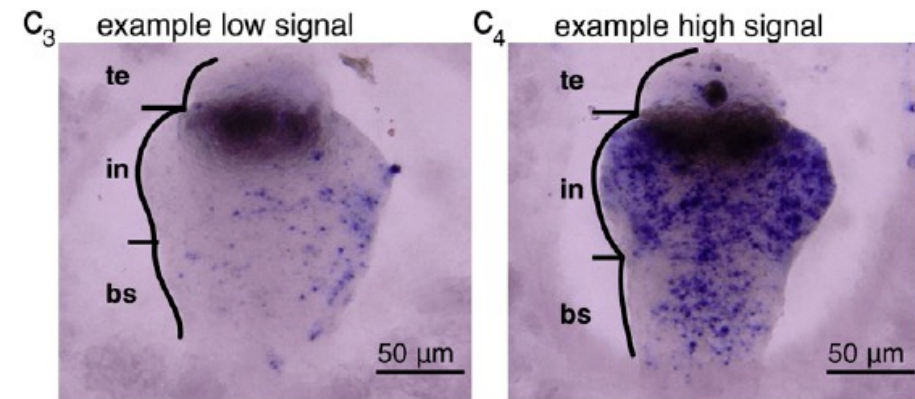
Abrams and colleagues identified **Chrnl-E**, one of several *Cassiopea* proteins related to nicotinic acetylcholine-receptor alpha subunits. A **ligand-gated ion channel** couples binding of a chemical ligand to opening of a pore through which ions cross the cell membrane.

Sequence comparisons identified a conserved **Cys-loop**, a cysteine-associated loop characteristic of this receptor family, and candidate ligand-binding features. Published structural predictions also showed a charged vestibule resembling that of human alpha-7 receptors. The figure presents predictions based on protein sequence, rather than a measured membrane current.

An open cation-permeable pore could change neuronal excitability by allowing charge to enter or leave a cell according to electrochemical gradients. Calcium permeability and the endogenous ligand were not established for Chrnl-E. The study did not demonstrate acetylcholine synthesis in *Cassiopea*, so classical cholinergic transmission remains unconfirmed.



Abrams et al. (2025), Fig. 3(A). Published Chrnl-E sequence-derived structural predictions.



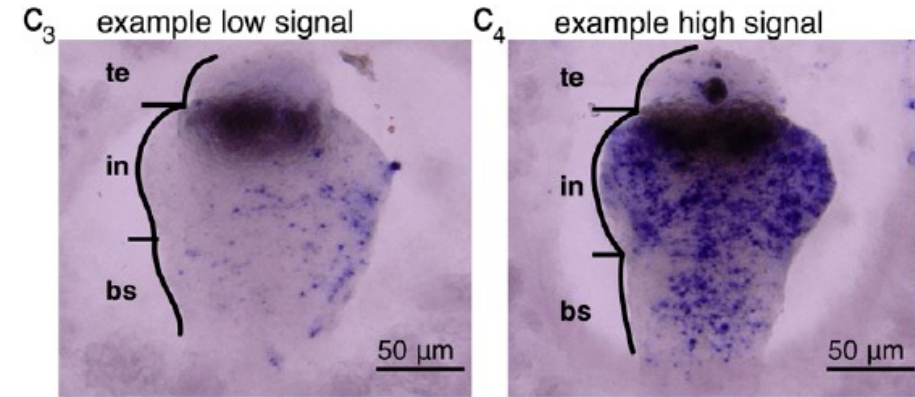
Abrams et al. (2025), Fig. 3(C,c3,c4). Low and high chrnal-E expression in rhopalial tissue.

# Candidate receptor RNA localizes to rhopalia

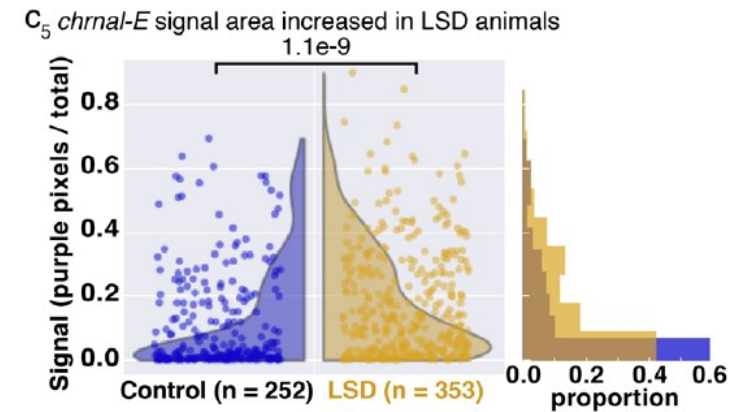
Abrams and colleagues used **in situ hybridization**, a method that detects a selected RNA sequence within tissue, to localize *chrnal-E* expression. Its labeled signal was concentrated in the neural region of *Cassiopea* rhopalia rather than distributed uniformly through the entire bell.

Both control and sleep-deprived animals contained rhopalia with stronger and weaker labeling. Sleep deprivation increased the labeled expression area in the measured preparations. Tissue localization linked a candidate receptor gene to organs already implicated in rhythm generation and sensory processing.

RNA localization identifies cells or tissue regions containing the transcript. It does not measure the amount of functional receptor protein at a membrane or identify all expressing cell types. Differences among rhopalia also caution against assuming that every marginal organ has an identical molecular state.



Abrams et al. (2025), Fig. 3(C,c3,c4). Low and high *chrnal-E* expression in rhopalian tissue.



Abrams et al. (2025), Fig. 3(C,c5). Quantified ganglionic *chrnal-E* expression area after sleep deprivation.

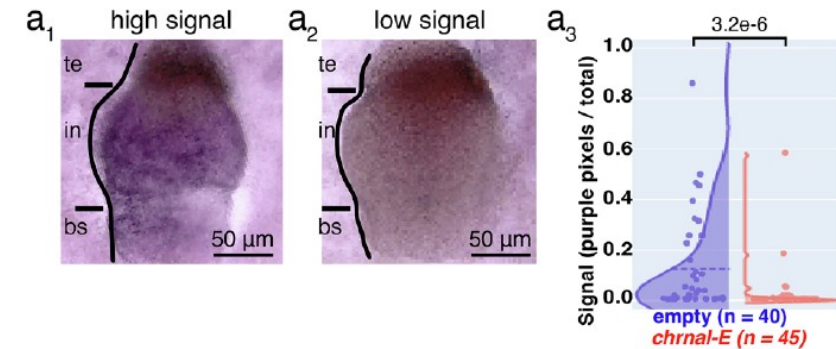
# RNA interference reduces a selected receptor transcript

**RNA interference**, or RNAi, uses double-stranded RNA to reduce expression of a matching gene. Abrams and colleagues fed *Cassiopea* material carrying a 387-base-pair *chrnal-E* sequence to test the behavioral contribution of this candidate receptor gene.

Animals received the RNAi treatment for two weeks before tissue-expression measurements. An empty-vector treatment controlled for the feeding and delivery procedure, while a separate minicollagen-targeting treatment tested whether another RNAi sequence produced the same phenotype. In situ labeling and quantitative RNA assays supported reduced *chrnal-E* expression.

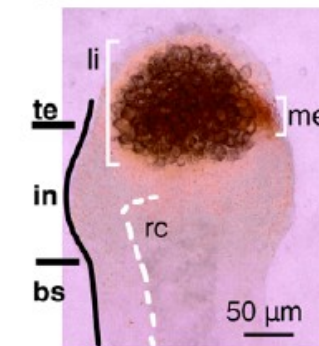
These controls strengthen the association between the targeted sequence and its molecular effect. Knockdown reduces expression rather than guaranteeing complete elimination in every cell. The manipulation therefore tests gene contribution at the organism and tissue levels without identifying the current carried by an individual receptor.

**A** RNAi knockdown of *chrnal-E* reduces expression in rhopalia



Abrams et al. (2025), Fig. 4(A). Rhopalian expression staining and *chrnal-E* knockdown measurements.

**C<sub>2</sub>** brightfield



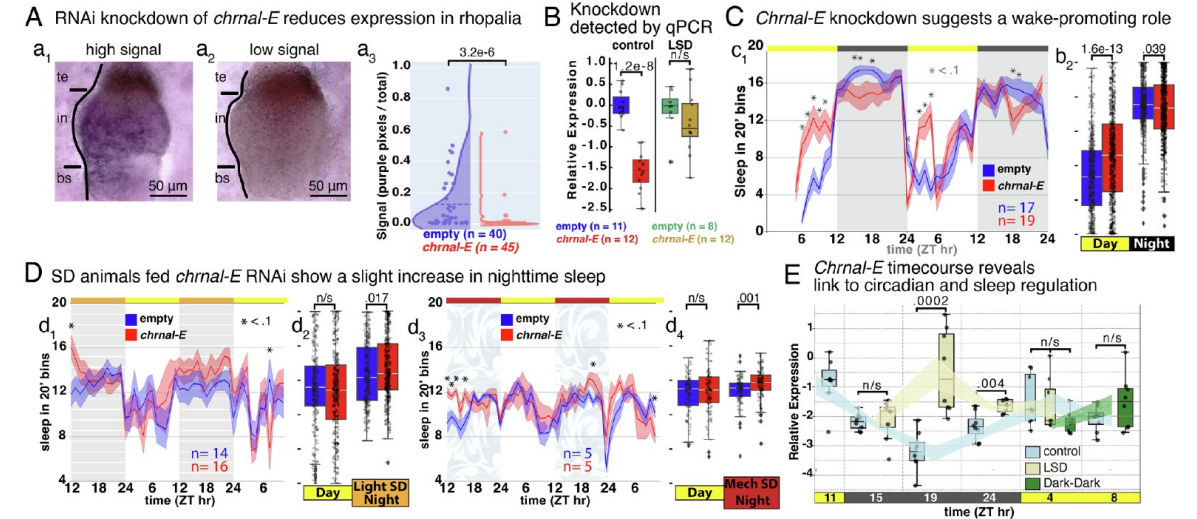
Abrams et al. (2025), Fig. 3(C,c2). *Cassiopea* rhopalium with terminal, intermediate and basal regions.

# Receptor knockdown redistributes daily sleep

Abrams and colleagues measured *Cassiopea* activity after *chrnal-E* RNAi and the corresponding control treatments. Their behavioral classification used prolonged pulse intervals together with arousal testing to distinguish a sleep-associated state from continuously active pulsing.

After two weeks of treatment, *chrnal-E* knockdown increased daytime sleep and reduced nighttime sleep. Empty-vector and minicollagen treatments supplied procedural and alternative-target comparisons. The phenotype was a redistribution of sleep across the daily cycle rather than evidence that all movement was abolished.

Gene perturbation links the candidate receptor to temporal regulation of behavioral state. It does not establish that *Chrnal-E* alone generates the sleep rhythm, or that every effect acts directly within pacemaker neurons. Developmental, sensory and network-level contributions remain possible within the targeted animals.



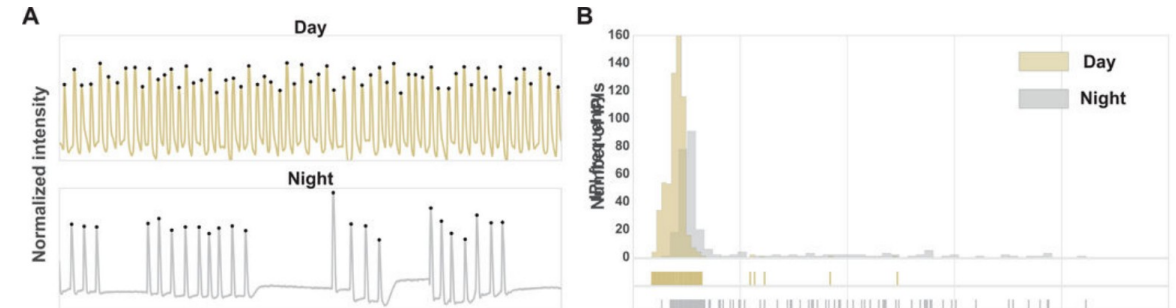
Abrams et al. (2025), Fig. 4(A–E). Expression and behavioral effects of *chrnal-E* knockdown, with controls.

# Bell-pulse timing changes between day and night

Nath and colleagues monitored *Cassiopea* under 12-hour light and 12-hour dark cycles, using video intensity changes to detect bell contractions. Animals were maintained at 26–28°C in seawater of 30–34 parts per thousand salinity, permitting repeated observation without handling each pulse.

Mean activity was 1,155 pulses per 20 minutes during the day and 781 pulses per 20 minutes at night, a reduction of approximately 32%. Night recordings included longer interpulse intervals and extended pauses. An **interpulse interval** is the time separating two detected bell contractions.

These measurements establish reversible daily changes in motor output. Reduced pulse rate alone is insufficient to identify sleep, because feeding, temperature or tissue impairment could also slow contractions. The study therefore combined activity measurements with tests of responsiveness and recovery after prolonged disturbance.



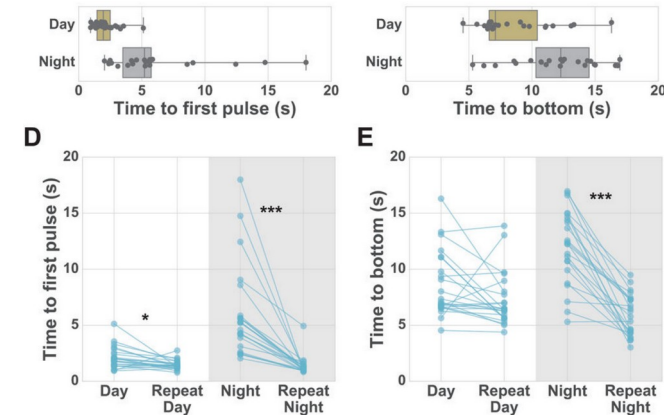
Nath et al. (2017), Fig. 2(A,B). Day and night pulse trains and inter-pulse-interval distributions.

# Nighttime quiescence includes a reversible arousal delay

To test **arousal**, the transition from quiescence to responsive activity, Nath and colleagues briefly supported *Cassiopea* above the bottom and then removed the support. The drop elicited contractions that returned the animals toward their normal resting position.

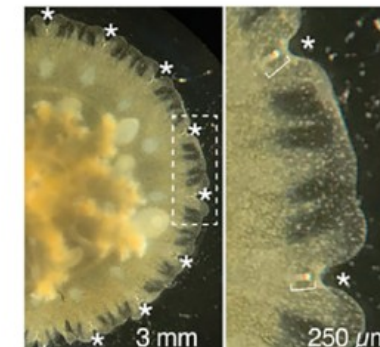
The first pulse occurred after about 2.1 seconds during the day and 5.9 seconds at night. When the drop was repeated within 30 seconds, the day–night difference disappeared. The repeated stimulus therefore tested whether the slower initial response reflected an inability to contract.

The delay followed by rapid recovery supports a reversible change in behavioral state. It does not identify which sensory cells detect the drop or which rhopalium initiates the first contraction. The test distinguishes quiescence from permanent motor failure while leaving the intervening cellular pathway to be resolved.



Nath et al. (2017), Fig. 3(B–E). First-pulse latency, return to bottom and repeated-drop controls.

## C Radial rhopalia (ganglia)



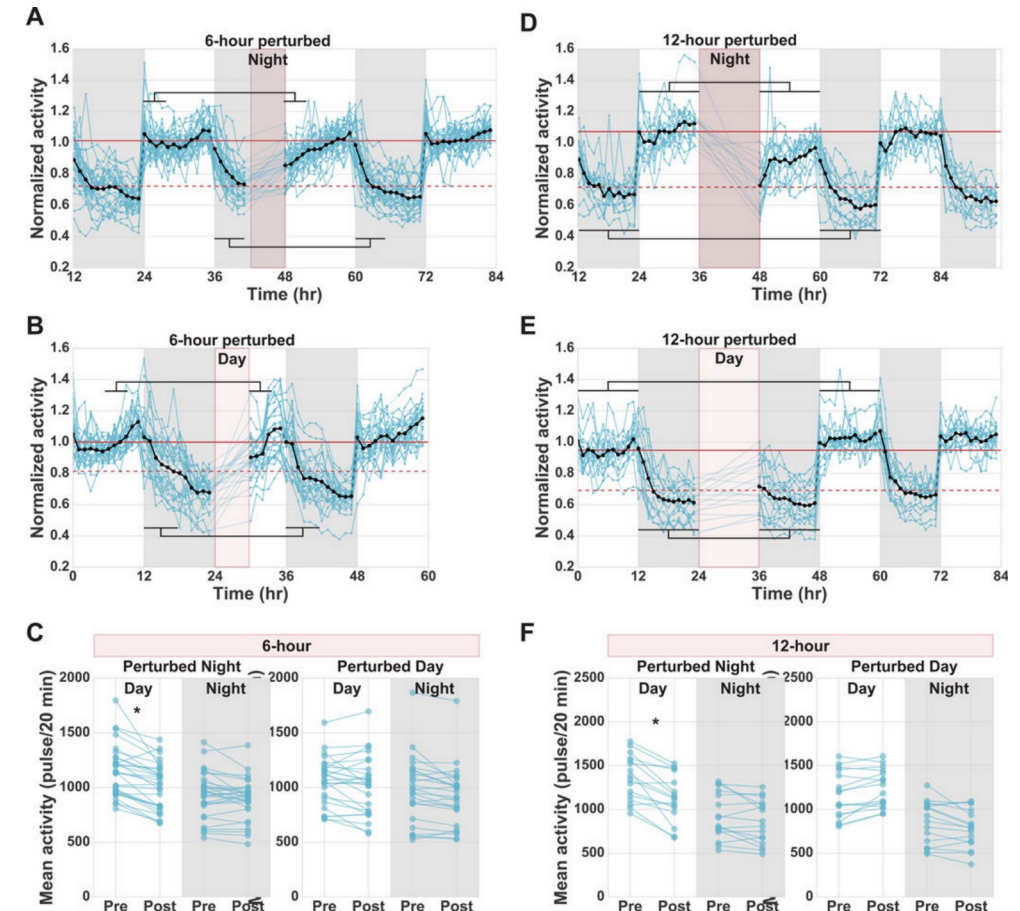
Abrams et al. (2025), Fig. 1(C). Radial distribution of rhopalia and a higher-magnification rhopalium.

# Nighttime disturbance produces compensatory quiescence

Nath and colleagues tested **sleep homeostasis**, the compensatory regulation of sleep after its disruption, by delivering water pulses for 10 seconds every 20 minutes. The stimulation interrupted nighttime quiescence for either 6 or 12 hours while video recorded the resulting activity.

Following the 6-hour treatment, pulse activity was about 12% lower during the early subsequent day; after 12 hours of disturbance, daytime activity was about 17% lower. Delivering comparable disturbance during the day did not produce the same subsequent reduction.

The timing control argues against a simple explanation based on mechanical exertion alone. Compensatory quiescence supports a homeostatically regulated sleep-like state, but pulse counts do not locate its cellular regulator. The study did not directly measure a molecular sleep-pressure signal or its action on a particular pacemaker cell.



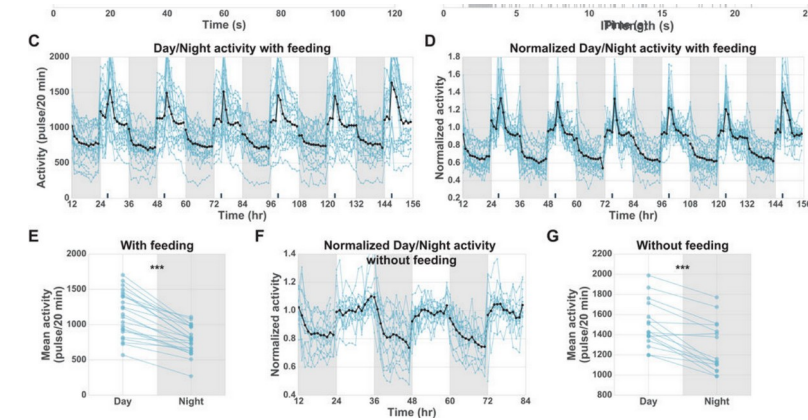
Nath et al. (2017), Fig. 4(A–F). Nighttime and daytime water-pulse perturbation with recovery recordings.

# Feeding reverses quiescence without explaining its rhythm

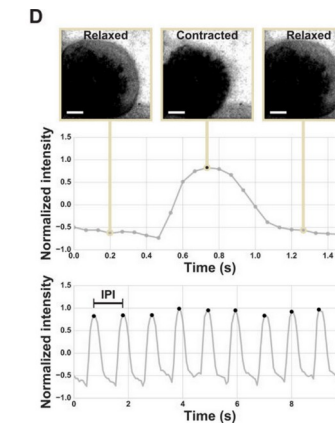
Cassiopea contractions contribute to feeding and fluid movement, so Nath and colleagues tested whether food availability explained the daily pulse rhythm. Feeding transiently increased pulse rate, and a day–night activity difference remained during three days without feeding.

Providing food at night reversed reduced activity, showing that quiescent animals retained a capacity to increase contractions in response to an ecologically relevant stimulus. The researchers also compared activity under constant darkness and constant light after prior light–dark entrainment.

Cycling that persisted in constant darkness was consistent with an endogenous timing contribution, whereas constant light disrupted the pattern. These results separate food-induced activation from the broader daily state change. They do not identify a circadian clock gene or prove that the same clock resides in every rhopalium.



Nath et al. (2017), Fig. 2(C–G). Long-term pulsation with and without daily feeding.



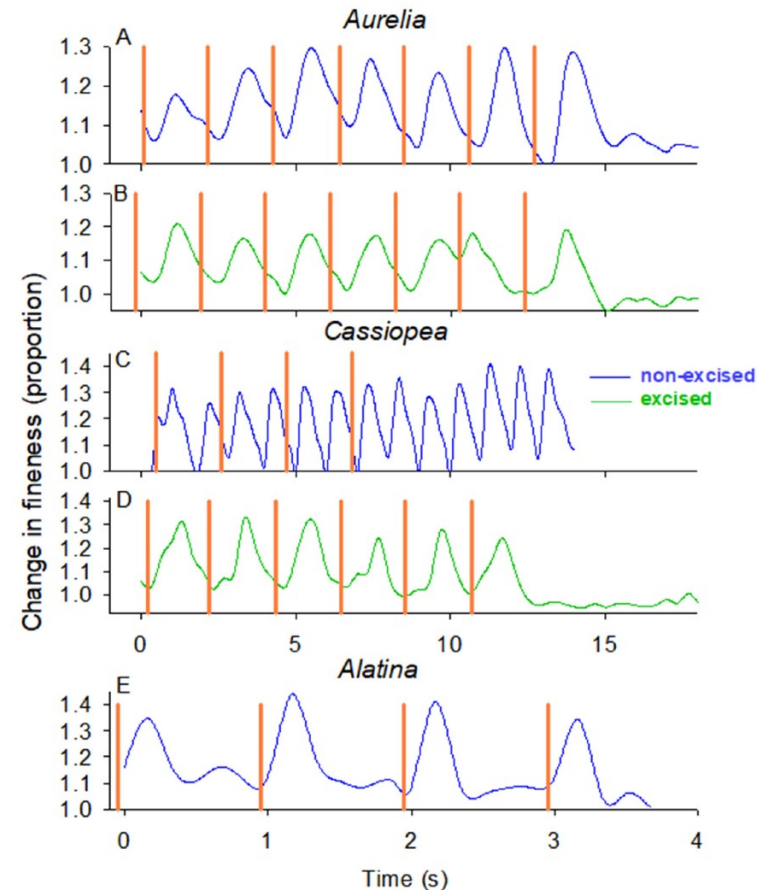
Nath et al. (2017), Fig. 1(D). Bell contractions and the measured inter-pulse interval.

# External stimulation competes with endogenous rhythms

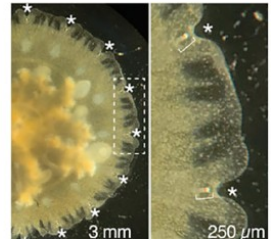
Anuszczyk and colleagues stimulated the subumbrellar tissue of adult *Cassiopea xamachana* using implanted electrodes and a lightweight controller. **Entrainment** here means contractions adopting the timing of an externally imposed pulse train, rather than the alignment of a daily clock.

During field trials, 0.5-Hz stimulation accompanied contractions near 1 Hz in intact *Cassiopea*. Contractions also continued after the external train stopped. The apparatus marked delivered electrical pulses with an LED so video could compare stimulation times with contraction events.

The intact rhythm was therefore not determined simply by the external command frequency. Endogenous activity can interfere with an imposed motor rhythm. This experiment measures the combined output of the animal and controller; it does not record competing signals from each rhopalium or establish a complete pacemaker-resetting rule.



C Radial rhopalia (ganglia)



Abrams et al. (2025), Fig. 1(C). Radial distribution of rhopalia and a higher-magnification rhopalium.

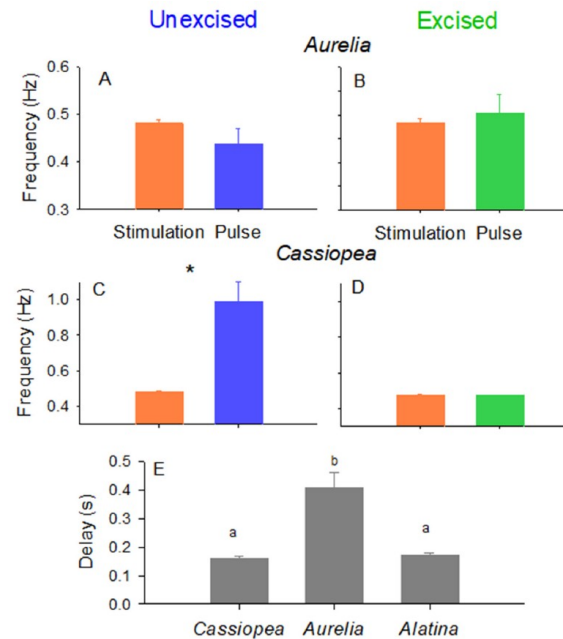
Anuszczyk et al. (2025), Fig. 2(A–E). Original intact and excised jellyfish recordings with stimulation times.

# Rhopalia removal improves control of pulse timing

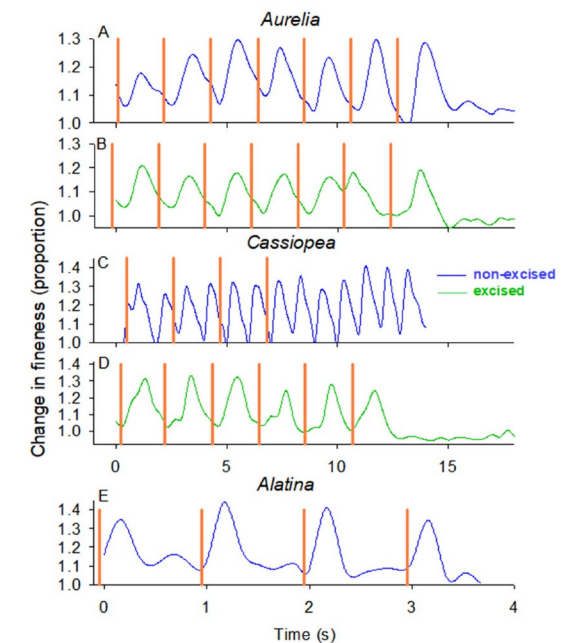
Anuszczyk and colleagues removed all rhopalia through small cuts at the bell margin, then tested external stimulation within two days to limit the opportunity for regeneration. This **ablation** physically removes a structure and asks whether its presence contributes to a measured function.

After removal, *Cassiopea xamachana* contracted at approximately 0.47 Hz during 0.5-Hz stimulation and stopped pulsing when stimulation ended. Intact animals maintained additional contractions. Externally evoked contractions demonstrated that bell tissue retained a functional path from stimulation to muscle movement after surgery.

The comparison strongly associates rhopalia with the endogenous activity that resists external timing control. Surgical injury is broader than a selective manipulation of one cell type. The result separates loss of spontaneous drive from total muscle paralysis without resolving the complete neural pathway activated by the electrodes.



Anuszczyk et al. (2025), Fig. 3(A–E). Original species comparison of pulse frequency and contraction delay.



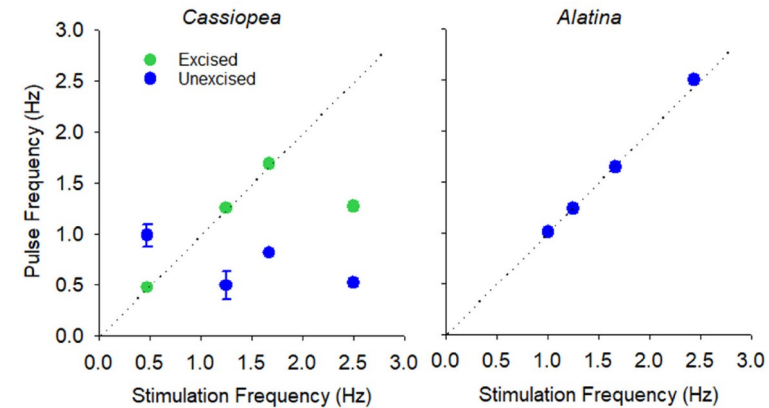
Anuszczyk et al. (2025), Fig. 2(A–E). Original intact and excised jellyfish recordings with stimulation times.

# Bell contractions have a finite following frequency

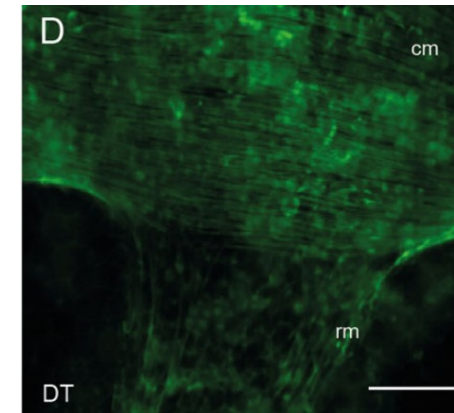
Anuszczyk and colleagues varied electrical-stimulation frequency in laboratory *Cassiopea andromeda* and the box jellyfish *Alatina alata*. These laboratory trials used a different *Cassiopea* species from the field trials, an important boundary when comparing performance across experiments.

*Cassiopea andromeda* did not follow stimulation above approximately 1.67 Hz, whereas *Alatina* followed rates up to 2.5 Hz. Removing rhopalia improved correspondence between imposed and observed frequencies within the tested range. Increasing input frequency therefore did not guarantee a proportional increase in completed contractions.

The upper following rate constrains the integrated neuromuscular system under these conditions. It could reflect excitation, muscle recovery or bell mechanics, which this assay did not isolate. The phalloidin-labeled *Cassiopea* musculature provides anatomical context rather than proof of the specific limiting process.



Anuszczyk et al. (2025), Fig. 5. Published stimulation-frequency versus pulse-frequency results.



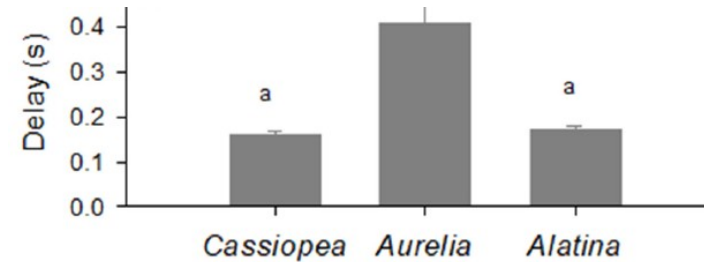
Muffett et al. (2025), Fig. 5(D). Outer circular fibers and radial lappet musculature.

# Stimulus-to-contraction latency combines several processes

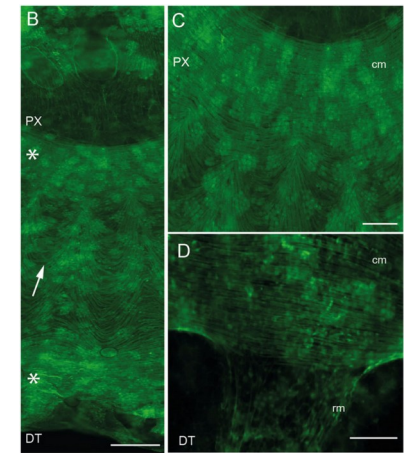
Anuszczyk and colleagues measured the time from delivered stimulation to the onset of contraction in video recordings. This **response latency** includes events between electrode activation and a visually detectable change in bell shape, rather than directly measuring nerve conduction alone.

*Cassiopea xamachana* began contracting after approximately 0.16 seconds, compared with 0.41 seconds in *Aurelia aurita* and 0.17 seconds in *Alatina alata*. Comparable stimulation and video detection linked input timing to the beginning of mechanical output across the tested animals.

Differences constrain the complete response pathway, including neural recruitment, excitation of muscle and tissue deformation. They cannot be converted into axonal conduction velocities without measured propagation distances and isolated neural timing. Anatomical muscle images identify the effector tissue but do not apportion the measured delay among these processes.



Anuszczyk et al. (2025), Fig. 3(E). Delay from electrical stimulus to bell contraction in three species.



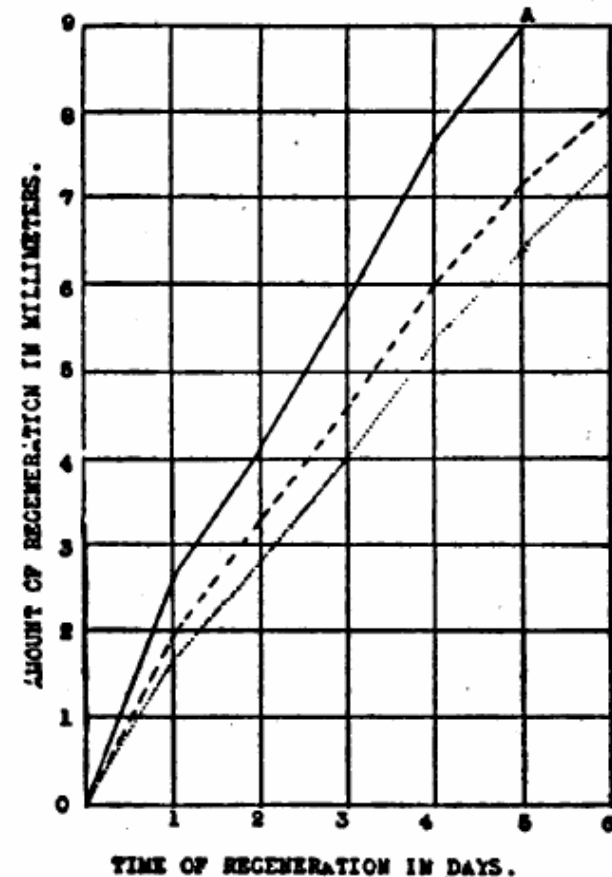
Muffett et al. (2025), Fig. 5(B–D). Actin-stained coronal muscle band and radial fibers.

# A closed conducting path can sustain a traveling wave

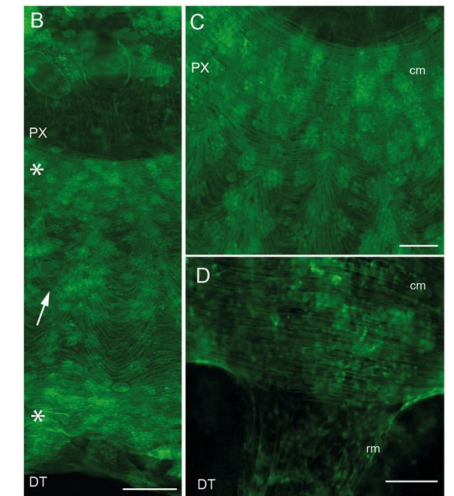
Mayer and Cary studied strips and ring-like preparations of *Cassiopea* subumbrellar tissue after removing marginal sense organs. The tissue could remain quiescent without rhopalia yet conduct an electrically initiated contraction along a connected path through the bell tissue.

In suitable closed circuits, an initiated wave continued circulating and produced repeated contractions without an intact marginal pacemaker. Cary used electrically activated circuit preparations to compare regeneration in active and inactive regions. The preparation separated wave initiation from continued travel along a circuit.

This **reentry** mechanism is repeated circulation of activity through a connected pathway. It shows that ongoing rhythmic contractions need not always imply a retained pacemaker organ. It does not establish reentry as the normal source of intact *Cassiopea* rhythms, nor does the historical preparation resolve individual synaptic connections.



Cary (1915), Fig. 1. Original comparison of regeneration with rhopalia, circuit activation or inactivity.



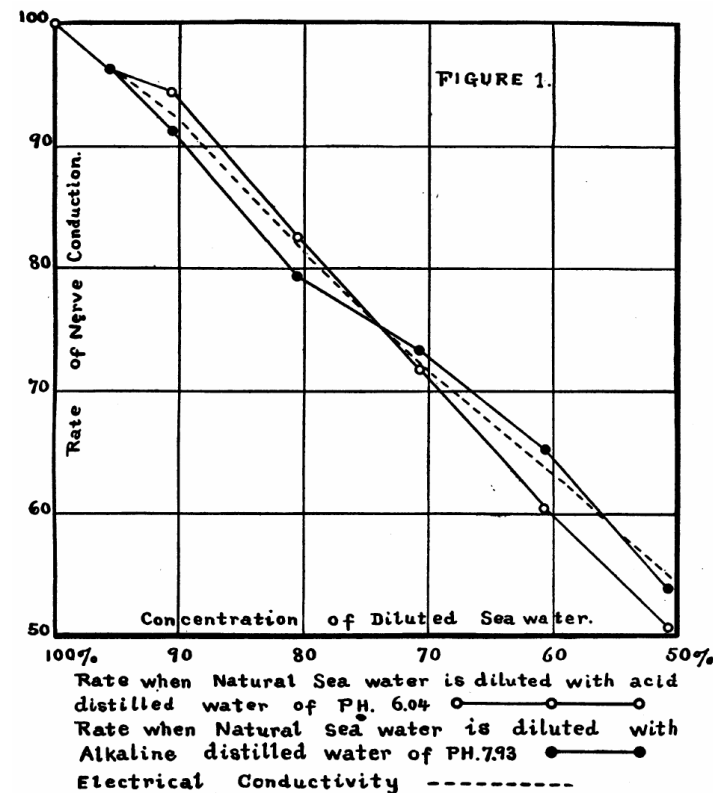
Muffett et al. (2025), Fig. 5(B–D). Actin-stained coronal muscle band and radial fibers.

# Ion-dilution experiments require matched chemical controls

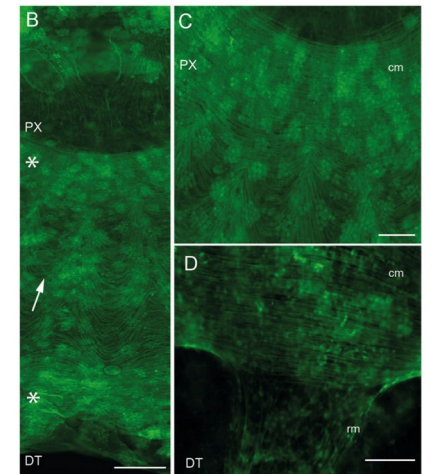
Mayer changed the seawater surrounding *Cassiopea* conducting preparations to test the dependence of wave propagation on ionic conditions. Dilution changes several variables together, including ion concentration, osmotic conditions and electrical conductivity of the extracellular fluid.

Experiments compared water dilution with dextrose and magnesium-salt treatments intended to separate these factors. The later study also controlled acidity: at 50% seawater concentration, propagation speed was about 54% of its value in normal seawater under the reported alkaline conditions.

Dependence on bath composition demonstrates that propagation is sensitive to the chemical environment. It does not identify the channels or synapses responsible. Mayer's initial adsorption-based explanation was explicitly revised in the 1916 paper after acidity and temperature effects were recognized; that earlier model should not be treated as an established modern mechanism.



Mayer (1916), Fig. 1. Conduction rate and bath conductivity under controlled-pH dilution.



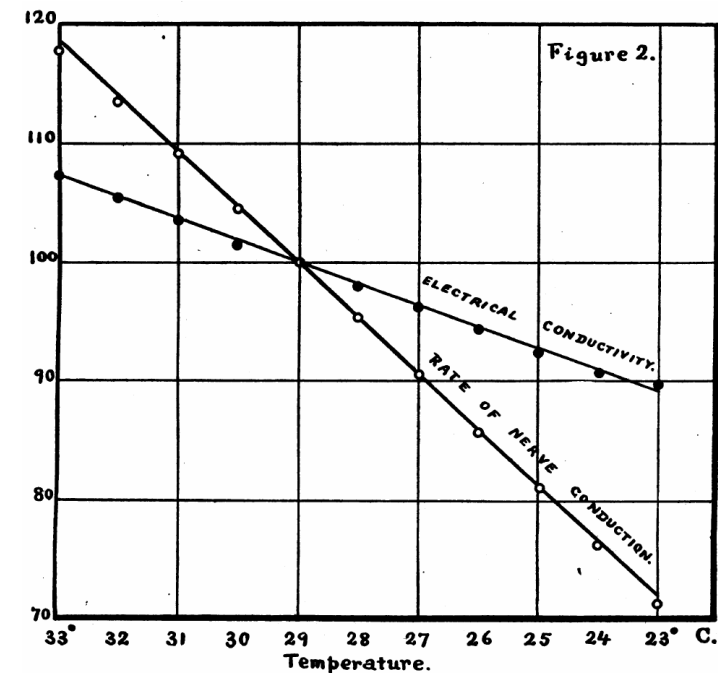
Muffett et al. (2025), Fig. 5(B–D). Actin-stained coronal muscle band and radial fibers.

# Temperature strongly affects wave propagation

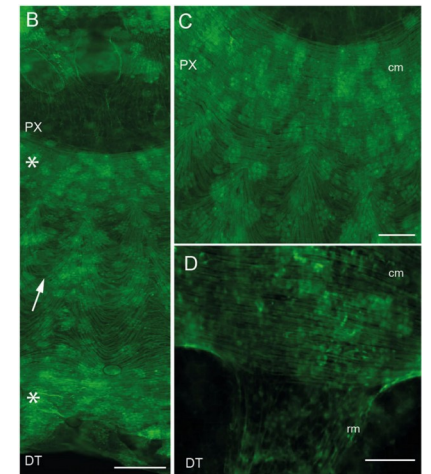
Mayer measured propagation in *Cassiopea* conducting tissue at different temperatures and compared the result with electrical conductivity of the surrounding seawater. These paired measurements tested whether changes in the extracellular medium alone could explain the tissue response.

Taking 29°C as 100%, propagation speed was approximately 71% at 23°C and 118% at 33°C. Bath conductivity changed less over the same temperature range. The nerve–muscle preparation therefore had a temperature dependence exceeding that of the external fluid.

The difference argues against explaining propagation entirely through bulk seawater conductivity. Temperature can influence membrane proteins, synaptic processes and muscle excitation, but the experiment did not isolate those components. The reported propagation measure should not be assigned to a specific modern ion channel without further physiological evidence.



Mayer (1916), Fig. 2. Conduction and bath-conductivity dependence on temperature.



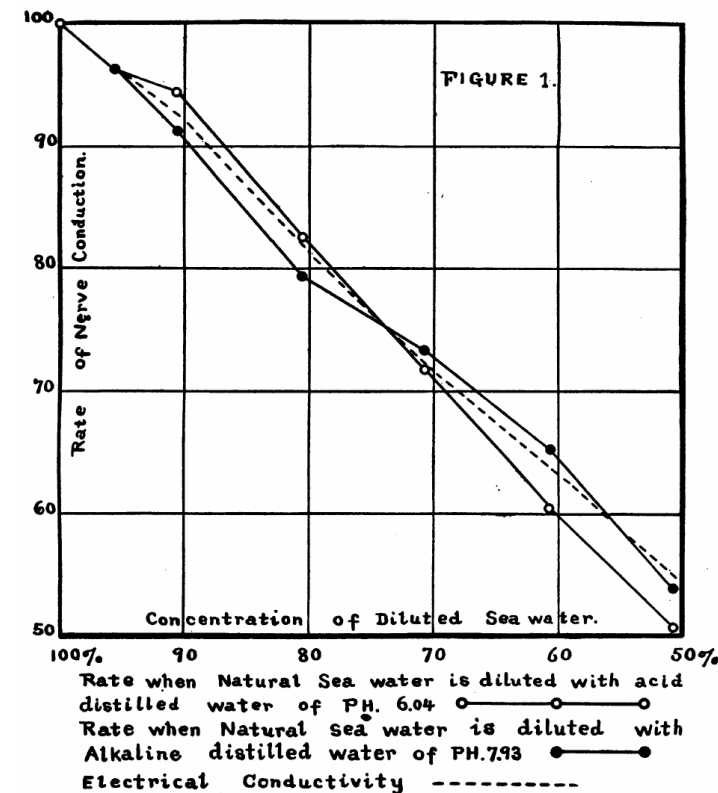
Muffett et al. (2025), Fig. 5(B–D). Actin-stained coronal muscle band and radial fibers.

# Equal bath conductivity does not ensure equal propagation

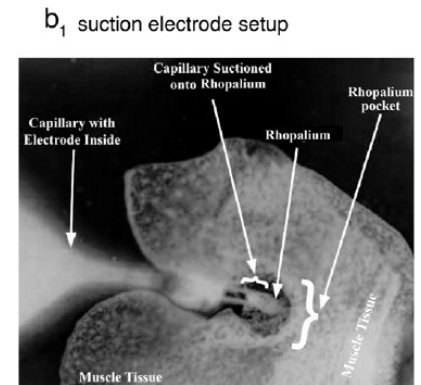
Mayer compared diluted seawater preparations with solutions containing magnesium salts or dextrose. The manipulations tested whether reduced propagation could be attributed solely to dilution of the bath, rather than to a change in the physiological requirements of the conducting tissue.

A 0.415-M magnesium-chloride dilution series could maintain substantial extracellular conductivity while tissue propagation declined. Dextrose-based comparisons also altered the relation between osmotic support and ionic composition. The observed tissue response therefore could not be inferred from a single bulk property of the surrounding solution.

These controls establish a physiological dependence on ionic conditions without resolving the molecular conductance. Modern rhopalial pharmacology addresses another question: regulation of local event generation. Neither result permits the untested conclusion that one receptor accounts for all wave propagation across the bell.



Mayer (1916), Fig. 1. Conduction rate and bath conductivity under controlled-pH dilution.



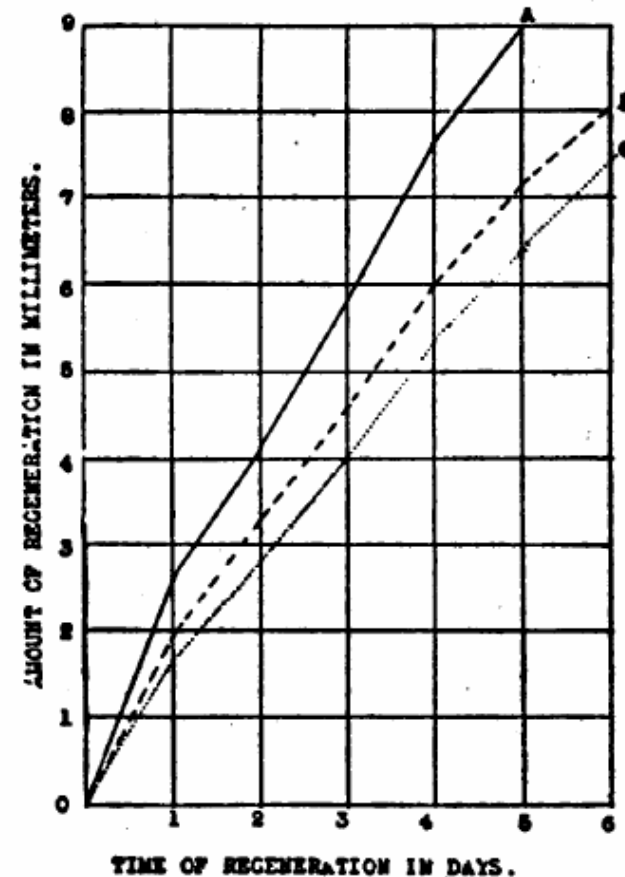
Abrams et al. (2025), Fig. 3(B,b1). Original suction-electrode recording preparation.

# Marginal organs affect regeneration beyond pulse count

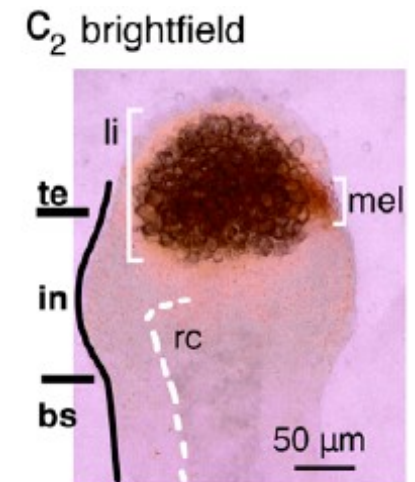
Cary compared two halves of the same *Cassiopea* bell, retaining rhopalia on one half and removing them from the other. Removing a comparable amount of tissue between organs on the retained side supplied a wound control while the two halves shared the same seawater environment.

On the first day, tissue regeneration was about 1.53 times faster on the side retaining sense organs. A separate electrically activated circuit preparation produced more contractions yet less regeneration than the rhopalia-retaining region. Comparable activity therefore did not reproduce the full organ-associated effect.

The result associates marginal sense organs with recovery beyond their contribution to pulse number alone. It does not identify a secreted trophic factor, a transmitter or a specific regenerative neuron. The historical controls reduce simple explanations based on tissue loss and movement while leaving the cellular mechanism unresolved.



Cary (1915), Fig. 1. Original comparison of regeneration with rhopalia, circuit activation or inactivity.



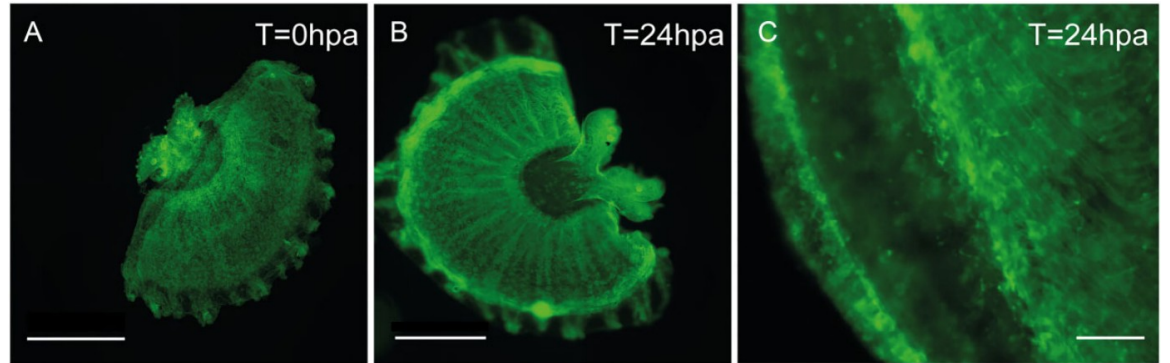
Abrams et al. (2025), Fig. 3(C,c2). *Cassiopea* rhopalium with terminal, intermediate and basal regions.

# Actin-rich tissue reorganizes near an ephyra wound

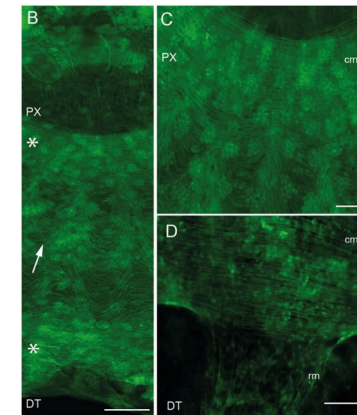
Muffett and colleagues bisected 5-mm *Cassiopea xamachana* ephyrae and examined muscle-associated actin at 0, 24, 48 and 72 hours after injury. Fluorescent **phalloidin**, a molecule binding filamentous actin, revealed the contractile tissue surrounding the cut margin.

At 24 hours, phalloidin signal was stronger near the injury than in the surrounding tissue. The authors observed actin-rich structures along the wound region as fragments changed shape. The micrographs preserve 1-mm overview and 50- $\mu$ m detail scale bars, distinguishing organ-scale remodeling from local tissue organization.

The staining supports early reorganization of actin-rich tissue during recovery. Increased signal is not a direct count of new muscle cells, and it does not demonstrate neural regeneration. Relating the new geometry to pacemaker coordination requires functional recordings in addition to the anatomical measurements.



Muffett et al. (2025), Fig. 6(A–C). Actin-stained fragments immediately and 24 h after bisection.



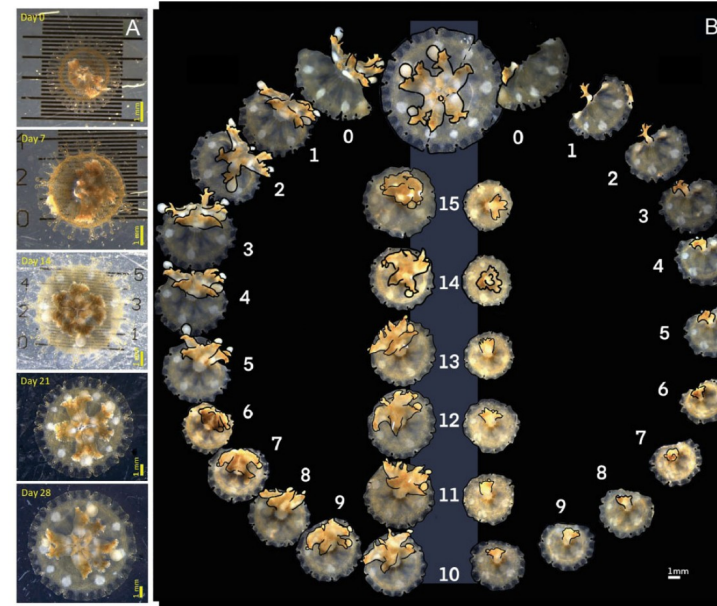
Muffett et al. (2025), Fig. 5(B–D). Actin-stained coronal muscle band and radial fibers.

# Halved ephyrae recover with fewer marginal organs

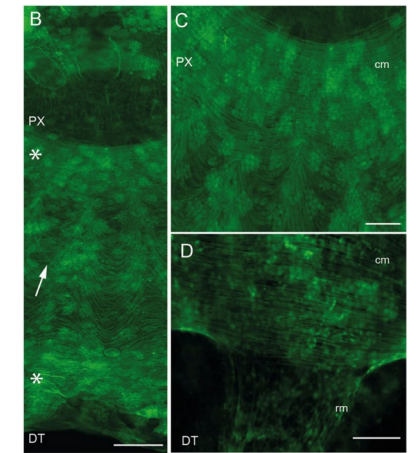
Muffett and colleagues followed *Cassiopea xamachana* ephyrae bisected at several ages after release. The cut divided both bell and manubrium, while uncut animals underwent comparable photography and handling. Repeated images tracked wound closure, growth and the eventual geometry of each fragment.

The mean wound angle decreased from approximately  $195^\circ$  to  $146.5^\circ$  during the first 24 hours. Over about two weeks, fragments became smaller functional ephyrae that pulsed and fed, retaining fewer rhopalia and oral arms than the original animal. Recovery did not require restoration of the original organ count.

The outcome demonstrates substantial functional tolerance to loss and remodeling in this life stage. It does not establish unchanged pulse coordination or complete reconstruction of the original neural network. Distributed bell tissue and surviving marginal organs provide a framework for recovery, while the precise contributions of each remain experimentally separable.



Muffett et al. (2025), Fig. 1(A,B). Original intact development and daily regeneration of two fragments.



Muffett et al. (2025), Fig. 5(B–D). Actin-stained coronal muscle band and radial fibers.

# Key takeaways

**Distributed organization:** Cassiopea has stage-specific neural and muscular architectures; detailed motor- and diffuse-net maps from Aurelia provide comparative evidence rather than a complete Cassiopea circuit.

**Localized rhythmic drive:** Isolated rhopalia retain electrical activity, and rhopalia removal permits externally controlled contractions, separating endogenous initiation from preserved motor capacity.

**Generation and propagation:** A pacemaker can initiate repeated activity, whereas a closed conducting path can sustain reentry; repeated bell movement alone does not distinguish these mechanisms.

**Bounded molecular causation:** Nicotine, tubocurarine and chrnl-E RNAi connect receptor-related mechanisms to rhythm and behavioral state, without establishing an endogenous acetylcholine transmitter pathway.

**State-dependent output:** Day–night pulse changes, reversible arousal delays and compensatory quiescence support regulated sleep-like behavior rather than simple muscle failure.

**Recovery with altered anatomy:** Rhopalia influence regeneration, and halved ephyrae regain pulsing and feeding with fewer marginal organs; functional recovery does not imply restoration of every original connection.